unit5mcq

1.

$$2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$$

A sample of N_2O_5 was placed in an evacuated container, and the reaction represented above occurred. The value of $P_{N_2O_5}$, the partial pressure of $N_2O_5(g)$, was measured during the reaction and recorded in the table below.

Time (min)	$P_{N_2O_5}$	$\ln(P_{N_2O_5})$	$\frac{1}{P_{N_2O_5}}$ (atm^{-1})
0	150	5.0	0.0067
100	75	4.3	0.013
200	38	3.6	0.027
300	19	2.9	0.053

Which of the following correctly describes the reaction?

(A) The decomposition of N_2O_5 is a zero-order reaction.

(B) The decomposition of N_2O_5 is a first-order reaction. ✓

(C) The decomposition of N_2O_5 is a second-order reaction.

(D) The overall reaction order is 3.

2. $2\ N_2O_5(g) \rightarrow 4\ NO_2(g) + O_2(g)$

A sample of N_2O_5 was placed in an evacuated container, and the reaction represented above occurred. The value of $P_{N_2O_5}$, the partial pressure of $N_2O_5(g)$, was measured during the reaction and recorded in the table below.

Time (min)	$P_{N_2O_5}$ (atm)	$\ln(P_{N_2O_5})$	$\frac{1}{P_{N_2O_5}}(atm^{-1})$
0	150	5.0	0.0067
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300	19	2.9	0.053

Which of the following correctly describes the reaction?

(A) The decomposition of N_2O_5 is a zero-order reaction.

(B) The decomposition of N_2O_5 is a first-order reaction. ✓

(C) The decomposition of N_2O_5 is a second-order reaction.

(D) The overall reaction order is 3.

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3. $2\ NOBr(g) \rightarrow 2\ NO(g) + Br_2(g)$

The equation above represents an elementary step in a chemical reaction. Which of the following is the correct expression for the rate law of the elementary step?

(A) Rate = $k[NOBr]^{½}$

(B) Rate = $k[NOBr]$

(C) Rate = $k[NOBr]^2$ ✓

(D) Rate = $k[NO]^2\ [Br_2]$

4.

$$5\ H_2O_2(aq) + 2\ MnO_4^-(aq) + 6\ H^+(aq) \rightarrow 2\ Mn^{2+}(aq) + 8\ H_2O(l) + 5\ O_2(g)$$

In a titration experiment, $H_2O_2(aq)$ reacts with aqueous $MnO_4^-(aq)$ as represented by the equation above. The dark purple $KMnO_4$ solution is added from a buret to a colorless, acidified solution of $H_2O_2(aq)$ in an Erlenmeyer flask. (Note: At the end point of the titration, the solution is a pale pink color.)

At a certain time during the titration, the rate of appearance of $O_2(g)$ was 1.0×10^{-3} mol/(L·s). What was the rate of disappearance of MnO_4^- at the same time?

(A) 6.0×10^{-3} mol/(L·s)

(B) 4.0×10^{-3} mol/(L·s)

(C) 6.0×10^{-4} mol/(L·s)

(D) 4.0×10^{-4} mol/(L·s) ✓

5. A kinetics experiment is set up to collect the gas that is generated when a sample of chalk, consisting primarily of solid $CaCO_3$, is added to a solution of ethanoic acid, CH_3COOH. The rate of reaction between $CaCO_3$ and CH_3COOH is determined by measuring the volume of gas generated at 25°C and 1 atm as a function of time. Which of the following experimental conditions is most likely to increase the rate of gas production?

(A) Decreasing the volume of ethanoic acid solution used in the experiment

(B) Decreasing the concentration of the ethanoic acid solution used in the experiment

(C) Decreasing the temperature at which the experiment is performed

(D) Decreasing the particle size of the $CaCO_3$ by grinding it into a fine powder ✓

6. A kinetics experiment is set up to collect the gas that is generated when a sample of chalk, consisting primarily of solid $CaCO_3$, is added to a solution of ethanoic acid, CH_3COOH. The rate of reaction between $CaCO_3$ and CH_3COOH is determined by measuring the volume of gas generated at 22°C and 1 atm as a function of time. Which of the following experimental conditions is most likely to increase the rate of gas production?

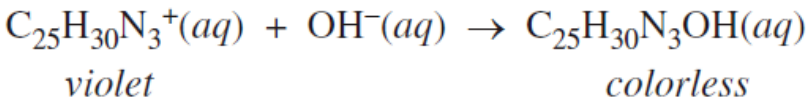
unit5mcq

(A) Decreasing the volume of ethanoic acid solution used in the experiment

(B) Decreasing the concentration of the ethanoic acid solution used in the experiment

(C) Decreasing the temperature at which the experiment is performed

(D) Decreasing the particle size of the $CaCO_3$ by grinding it into a fine powder ✓



The reaction between $C_{25}H_{30}N_3^+$ (*aq*) and OH^-(*aq*), as represented above, is first order with respect to $C_{25}H_{30}N_3^+$ (*aq*) in the presence of excess OH^-(*aq*). A 10.0 mL sample of 0.10 *M* NaOH(*aq*) is mixed with a 10.0 mL sample of 2.5 x 10^{-5} *M* $C_{25}H_{30}N_3^+$(*aq*). A 5.0 mL sample of the mixture is quickly transferred to a clean cuvette and placed in a spectrophotometer, and the progress of the reaction is measured. The data are given in the table below.

Time (s)	0	30	60	90	120	150	180	210	240	270	300
Absorbance	0.62	0.54	0.47	0.41	0.36	0.31	0.27	0.23	0.20	0.17	0.15

7. Approximately how long did it take for 75 percent of the initial amount of $C_{25}H_{30}N_3^+$ (*aq*) to react?

(A) 75 s

(B) 225 s

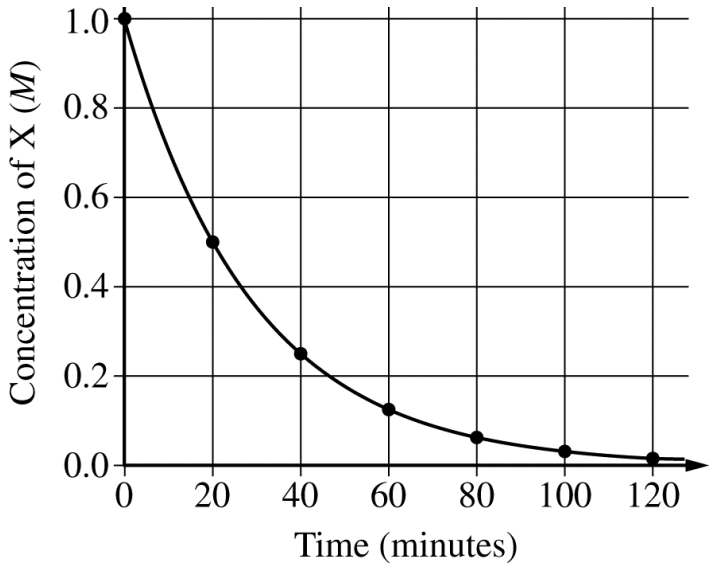
(C) 300 s ✓

(D) 600 s

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8. X → Products

A student studied the kinetics of the reaction represented above by measuring the concentration of the reactant, X, over time. The data are plotted in the graph below.



Which of the following procedures will allow the student to determine the rate constant, k , for the reaction?

(A) Plot ln [X] versus time and determine the magnitude of the slope. ✓

(B) Plot 1/[X] versus time and determine the magnitude of the slope.

(C) Run another trial of the experiment with a different initial concentration, plot the data on the same graph, and see where the curves intersect.

(D) Run another trial of the experiment at a different temperature, plot the data on the same graph, and see where the curves have the same slope.

9.

$$C_2H_4(g) + H_2(g) \rightarrow C_2H_6(g)$$

Which of the following will most likely increase the rate of the reaction represented above?

(A) Decreasing the temperature of the reaction system

(B) Adding a heterogeneous catalyst to the reaction system ✓

(C) Increasing the volume of the reaction vessel using a piston

(D) Removing some $H_2(g)$ from the reaction system

unit5mcq

10.

$$C_2H_4(g) + H_2(g) \xrightarrow{\text{platinum}} C_2H_6(g)$$

$C_2H_4(g)$ is reduced by $H_2(g)$ in the presence of a solid platinum catalyst, as represented by the equation above. Factors that could affect the rate of the reaction include which of the following?

I. Changes in the partial pressure of $H_2(g)$

II. Changes in the particle size of the platinum catalyst

III. Changes in the temperature of the reaction system

(A) III only

(B) I and II only

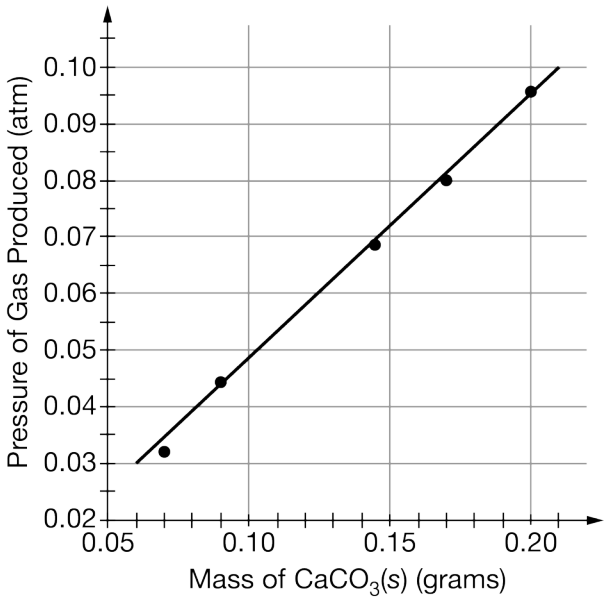
(C) I and III only

(D) II and III only

(E) I, II, and III ✓

unit5mcq

An experiment is performed to measure the mass percent of $\mathrm{CaCO_3}(s)$ in eggshells. Five different samples of $\mathrm{CaCO_3}(s)$ of known mass react with an excess of $2.0\ M\ \mathrm{HCl}(aq)$ in identical sealed, rigid reaction vessels. The pressure of the gas produced is measured with a pressure gauge attached to the reaction vessel. Since the reaction is exothermic, the reaction system is cooled to its original temperature before the pressure is recorded. The experimental data are used to create the calibration line below.



The experiment is repeated with an eggshell sample, and the experimental data are recorded in the table below.

Mass of eggshell sample	$0.200\ \mathrm{g}$
Pressure prior to reaction	$0.800\ \mathrm{atm}$
Pressure at completion of reaction	$0.870\ \mathrm{atm}$

11. Which of the following modifications will increase the rate of the reaction the most?

(A) Using $2.0\ M\ \mathrm{CH_3COOH}(aq)$ instead of $2.0\ M\ \mathrm{HCl}(aq)$

(B) Cooling the $\mathrm{HCl}(aq)$ to a lower temperature than it was in the original experiment

(C) Reducing the volume of the reaction vessel

(D) Using eggshells that are more finely powdered than those used in the original experiment ✓

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Step 1: $\mathrm{H_2}(g) + \mathrm{ICl}(g) \rightarrow \mathrm{HI}(g) + \mathrm{HCl}(g)$ (slow)

Step 2: $\mathrm{HI}(g) + \mathrm{ICl}(g) \rightarrow \mathrm{HCl}(g) + \mathrm{I_2}(g)$ (fast)

12. Which of the following represents the overall chemical equation for the reaction and the rate law for elementary step 2 ?

(A) The overall reaction is $\mathrm{H_2}(g) + 2\ \mathrm{ICl}(g) \rightarrow \mathrm{HCl}(g) + \mathrm{I_2}(g)$; the rate law for step 2 is $rate = k[\mathrm{H_2}][\mathrm{HI}][\mathrm{ICl}]$.

(B) The overall reaction is $\mathrm{H_2}(g) + 2\ \mathrm{ICl}(g) \rightarrow 2\ \mathrm{HCl}(g) + \mathrm{I_2}(g)$; the rate law for step 2 is $rate = k[\mathrm{H_2}][\mathrm{ICl}]^2$.

(C) The overall reaction is $\mathrm{H_2}(g) + \mathrm{ICl}(g) \rightarrow \mathrm{HCl}(g) + \mathrm{I_2}(g)$; the rate law for step 2 is $rate = k[\mathrm{H_2}][\mathrm{ICl}]$.

(D) The overall reaction is $\mathrm{H_2}(g) + 2\ \mathrm{ICl}(g) \rightarrow 2\ \mathrm{HCl}(g) + \mathrm{I_2}(g)$; the rate law for step 2 is $rate = k[\mathrm{HI}][\mathrm{ICl}]$. ✓

13. Which of the following represents a rate law for the overall reaction that is consistent with the proposed mechanism?

(A) $\mathrm{Rate} = k[\mathrm{H_2}][\mathrm{ICl}]$ ✓

(B) $\mathrm{Rate} = k[\mathrm{H_2}][\mathrm{ICl}]^2$

(C) $\mathrm{Rate} = k[\mathrm{H_2}][\mathrm{HI}][\mathrm{ICl}]^2$

(D) $\mathrm{Rate} = k[\mathrm{HI}][\mathrm{ICl}]$

14. The reaction is carried out at constant temperature inside a rigid container. Based on this mechanism, which of the following is the most likely reason for the different rates of step 1 and step 2 ?

(A) The only factor determining the rate of step 2 is the orientation of the HI and ICl polar molecules during a collision, but it has a negligible effect when $\mathrm{H_2}$ and ICl molecules collide.

(B) The amount of energy required for a successful collision between $\mathrm{H_2}$ and ICl is greater than the amount of energy required for a successful collision between HI and ICl. ✓

(C) The fraction of molecules with enough energy to overcome the activation energy barrier is lower for HI and ICl than for $\mathrm{H_2}$ and ICl.

(D) The frequency of collisions between $\mathrm{H_2}$ and ICl is greater than the frequency of collisions between HI and ICl.

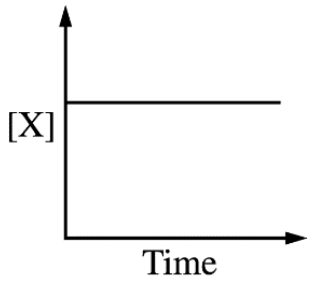
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15. X → products

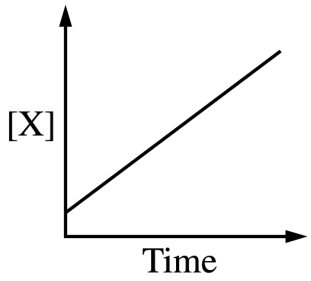
Pure substance X decomposes according to the equation above. Which of the following graphs indicates that the rate of decomposition is second order in X ?

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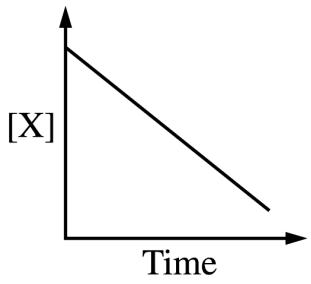
(A)



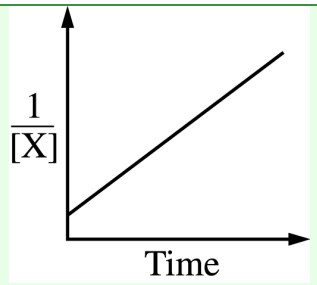
(B)



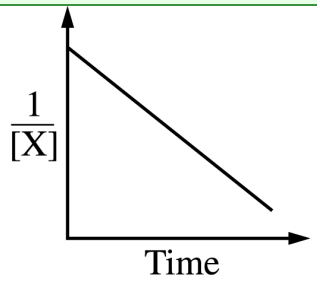
(C)



(D)



(E)



16. Two solid reactants of varying particle size are combined in a vessel. Assuming that the same mass of each reactant is used in each case, which of the following represents the conditions that will result in the fastest reaction rate?

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(A) 
Temperature = 950° C

(B) 
Temperature = 500° C

(C) 
Temperature = 500° C

(D) 
Temperature = 950° C ✓

$$Cu(s) + 4\ HNO_3(aq) \rightarrow Cu(NO_3)_2(aq) + 2\ NO_2(g) + 2\ H_2O(l)$$

Each student in a class placed a 2.00 g sample of a mixture of Cu and Al in a beaker and placed the beaker in a fume hood. The students slowly poured 15.0 mL of 15.8 *M* $HNO_3(aq)$ into their beakers. The reaction between the copper in the mixture and the $HNO_3(aq)$ is represented by the equation above. The students observed that a brown gas was released from the beakers and that the solutions turned blue, indicating the formation of $Cu^{2+}(aq)$. The solutions were then diluted with distilled water to known volumes.

17. In one student's experiment the reaction proceeded at a much slower rate than it did in the other students' experiments. Which of the following could explain the slower reaction rate?

(A) In the student's sample the metal pieces were much smaller than those in the other students' samples.

(B) The student heated the reaction mixture as the $HNO_3(aq)$ was added.

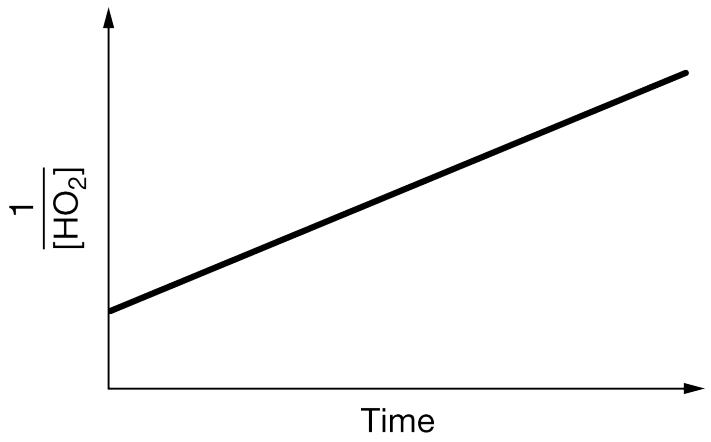
(C) The student used a 1.5 *M* solution of $HNO_3(aq)$ instead of a 15.8 *M* solution of $HNO_3(aq)$. ✓

(D) The student used a 3.00 g sample of the mixture instead of the 2.00 g sample that was used by the other students.

unit5mcq

18. $2\ HO_2(g) \rightarrow H_2O_2(g) + O_2(g)$

The reaction represented by the chemical equation shown above occurs in Earth's atmosphere. In an experiment, $[HO_2]$ was monitored over time and the data plotted as shown in the following graph.



Based on the information, which of the following is the rate law expression for the reaction?

(A) $\text{Rate} = k$

(B) $\text{Rate} = k[HO_2]$

(C) $\text{Rate} = k[HO_2]^2$ ✓

(D) $\text{Rate} = k[H_2O_2][O_2]$

19. After a certain pesticide compound is applied to crops, its decomposition is a first-order reaction with a half-life of 56 days. What is the rate constant, k, for the decomposition reaction?

(A) $0.012\ \text{day}^{-1}$ ✓

(B) $0.018\ \text{day}^{-1}$

(C) $56\ \text{day}^{-1}$

(D) $81\ \text{day}^{-1}$

20. After a certain pesticide compound is applied to crops, its decomposition is a first-order reaction with a half-life of 32.5 days. What is the rate constant, k, for the decomposition reaction?

(A) $0.0213\ \text{day}^{-1}$ ✓

(B) $0.0308\ \text{day}^{-1}$

(C) $32.5\ \text{day}^{-1}$

(D) $46.9\ \text{day}^{-1}$

unit5mcq

21.

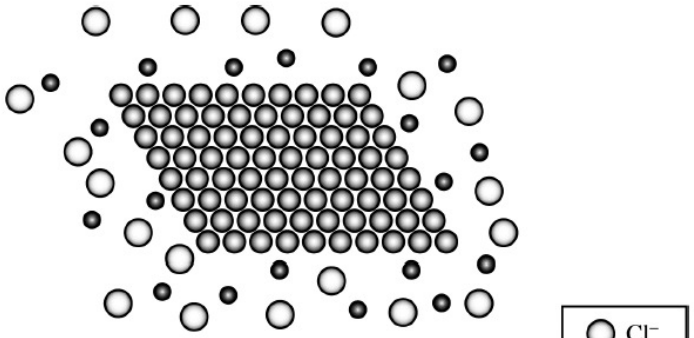


Figure 1

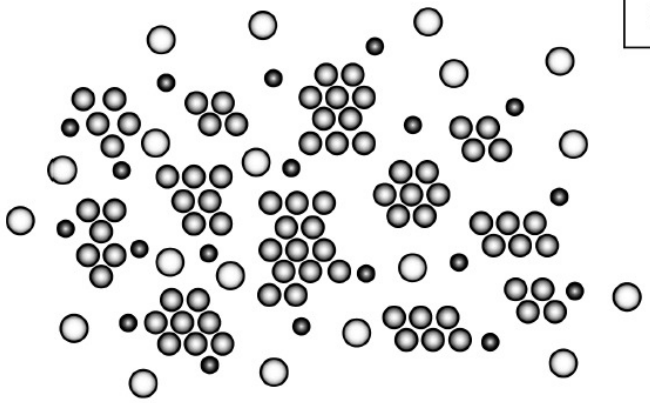


Figure 2

Two samples of Mg*(s)* of equal mass were placed in equal amounts of HCI*(aq)* contained in two separate reaction vessels. Particle representations of the mixing of Mg*(s)* and HCI*(aq)* in the two reaction vessels are shown in Figure 1 and Figure 2 above. Water molecules are not included in the particle representations. Which of the reactions will initially proceed faster, and why?

(A) The reaction in Figure 1, because the atoms of Mg are more concentrated than those in Figure 2

(B) The reaction in Figure 1, because the Mg*(s)* in Figure 1 has a larger mass than the Mg*(s)* in Figure 2

(C) The reaction in Figure 2, because more Mg atoms are exposed to HCI(*aq)* in Figure 2 than in Figure 1 ✓

(D) The reaction in Figure 2, because the Mg*(s)* in Figure 2 has less surface area than the Mg*(s)* in Figure 1

A 0.50 g sample of $\mathrm{Mg}(s)$ was placed in a solution of $\mathrm{HCl}(aq)$, where it reacted completely.

22. In a third experiment, 0.10 g samples of $\mathrm{Mg}(s)$ are placed in excess $\mathrm{HCl}(aq)$ of various concentrations: $0.050\ M$, $0.10\ M$, $0.25\ M$, and $0.50\ M$. The reactions are run in successive order from $0.050\ M$ to $0.50\ M$, and the time required for each reaction to go to completion is recorded. As the concentration of $\mathrm{HCl}(aq)$ increases from $0.050\ M$ to $0.50\ M$, which of the following is the expected result?

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(A) The reaction time increases, and the rate of the reaction decreases.

(B) The reaction time decreases, and the rate of the reaction increases. ✓

(C) Both the reaction time and the rate of the reaction increase.

(D) Both the reaction time and the rate of the reaction decrease.

23.

$$H_3AsO_4 + 3\ I^- + 2\ H3O^+ \rightarrow H_3AsO_3 + I_3^- + H_2O$$

The oxidation of iodide ions by arsenic acid in acidic aqueous solution occurs according to the stoichiometry shown above. The experimental rate law of the reaction is:

$$\text{Rate} = k[H_3AsO_4]\ [I^-]\ [H_3O^+]$$

According to the rate law for the reaction, an increase in the concentration of hydronium ion has what effect on this reaction?

(A) The rate of reaction increases. ✓

(B) The rate of reaction decreases.

(C) The value of the equilibrium constant increases.

(D) The value of the equilibrium constant decreases.

(E) Neither the rate nor the value of the equilibrium constant is changed.

24. Step 1: $Cl(g) + O_3(g) \rightarrow ClO(g) + O_2(g)$

Step 2: $ClO(g) + O(g) \rightarrow Cl(g) + O_2(g)$

A proposed mechanism for destruction of ozone gas in the stratosphere is represented above. Which of the following is evidence that the mechanism is occurring?

(A) The presence of Cl(*g*) increases the rate of the overall reaction. ✓

(B) The presence of Cl(*g*) decreases the rate of the overall reaction.

(C) The presence of Cl(*g*) increases the equilibrium constant of the overall reaction.

(D) The presence of Cl(*g*) decreases the equilibrium constant of the overall reaction.

25. If the oxygen isotope ^{20}O has a half-life of 15 seconds, what fraction of a sample of pure ^{20}O remains after 1.0 minute?

unit5mcq

(A) $\frac{1}{2}$

(B) $\frac{1}{4}$

(C) $\frac{7}{30}$

(D) $\frac{1}{8}$

(E) $\frac{1}{16}$ ✓

26. Gaseous cyclobutene undergoes a first-order reaction to form gaseous butadiene. At a particular temperature, the partial pressure of cyclobutene in the reaction vessel drops to one-eighth its original value in 124 seconds. What is the half-life for this reaction at this temperature?

(A) 15.5 sec

(B) 31.0 sec

(C) 41.3 sec ✓

(D) 62.0 sec

(E) 124 sec

27. The isomerization of cyclopropane to propylene is a first-order process with a half-life of 19 minutes at 500°C. The time it takes for the partial pressure of cyclopropane to decrease from 1.0 atmosphere to 0.125 atmosphere at 500°C is closest to

(A) 38 minutes

(B) 57 minutes ✓

(C) 76 minutes

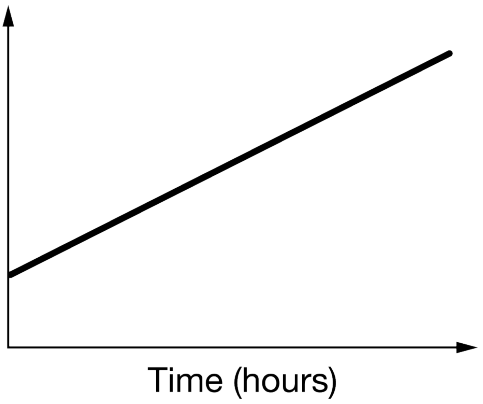
(D) 152 minutes

(E) 190 minutes

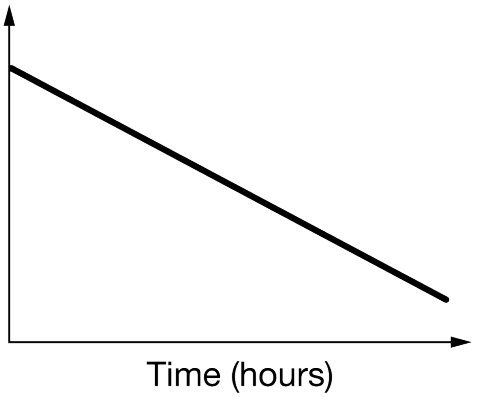
28. The rate constant (k) for the decay of the radioactive isotope I-131 is $3.6 \times 10^{-3}\ \text{hours}^{-1}$. The slope of which of the following graphs is correct for the decay and could be used to confirm the value of k ?

unit5mcq

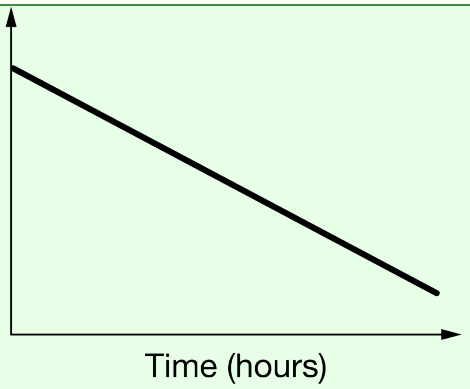
(A) $\frac{1}{[\text{I-131}]}$



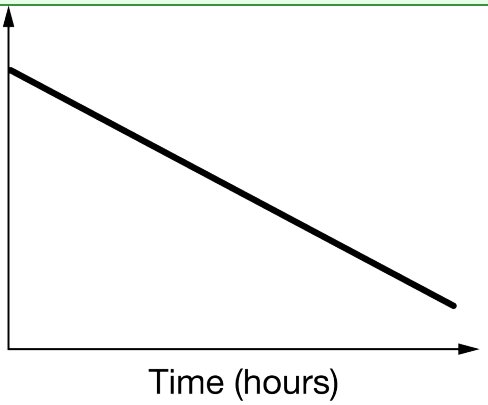
(B) $\frac{1}{[\text{I-131}]^2}$



(C) $\ln[\text{I-131}]$



(D) $[\text{I-131}]$



unit5mcq

$$2\ H_2O_2(aq) \rightarrow 2\ H_2O(l) + O_2(g) \qquad \Delta H° = -196\ \text{kJ/mol}_{rxn}$$

The decomposition of $H_2O_2(aq)$ is represented by the equation above. A student monitored the decomposition of a 1.0 L sample of $H_2O_2(aq)$ at a constant temperature of 300. K and recorded the concentration of H_2O_2 as a function of time. The results are given in the table below.

Time (s)	$[H_2O_2]$
0	2.7
200.	2.1
400.	1.7
600.	1.4

29. Which of the following statements is a correct interpretation of the data regarding how the order of the reaction can be determined?

(A) The reaction must be first order because there is only one reactant species.

(B) The reaction is first order if the plot of ln $[H_2O_2]$ versus time is a straight line. ✓

(C) The reaction is first order if the plot of $1/[H_2O_2]$ versus time is a straight line.

(D) The reaction is second order because 2 is the coefficient of H_2O_2 in the chemical equation.

30.

$$H_3AsO_4 + 3\ I^- + 2\ H3O^+ \rightarrow H_3AsO_3 + I_3^- + H_2O$$

The oxidation of iodide ions by arsenic acid in acidic aqueous solution occurs according to the stoichiometry shown above. The experimental rate law of the reaction is:

$$\text{Rate} = k[H_3AsO_4]\ [I^-]\ [H_3O^+]$$

What is the order of the reaction with respect to I^-?

(A) 1 ✓

(B) 2

(C) 3

(D) 5

(E) 6

unit5mcq

31. If 87.5 percent of a sample of pure ^{131}I decays in 24 days, what is the half-life of ^{131}I?

(A) 6 days

(B) 8 days ✓

(C) 12 days

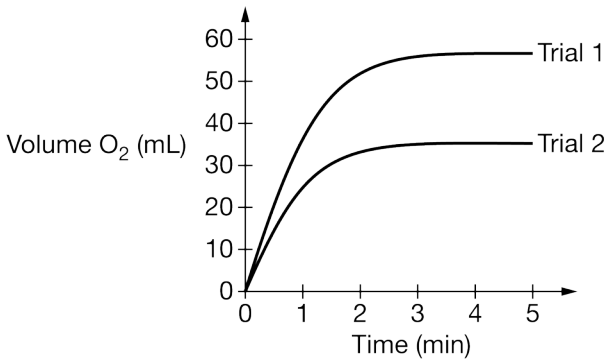
(D) 14 days

(E) 21 days

32. $2\ H_2O_2(aq) \xrightarrow{KI(s)} 2\ H_2O(l) + O_2(g)$

$\text{Rate} = k[H_2O_2]$

A student performed an experiment to study the factors that affect the rate of the first-order catalytic decomposition of $H_2O_2(aq)$, represented above.



The student monitored the volume of $O_2(g)$ produced over time as the reaction proceeded. The data from two trials are plotted in the graph above. Which of the following best explains the results of trial 2 compared with those of trial 1 ?

(A) The reaction temperature in trial 2 was higher than it was in trial 1.

(B) The volume of the reaction vessel was larger in trial 2 than it was in trial 1.

(C) The student used a smaller amount of $KI(s)$ in trial 2 compared with that used in trial 1.

(D) The concentration of $H_2O_2(aq)$ was lower in trial 2 than it was in trial 1. ✓

33. $2\ NO(g) + 2\ H_2(g) \rightarrow N_2(g) + 2\ H_2O(g)$

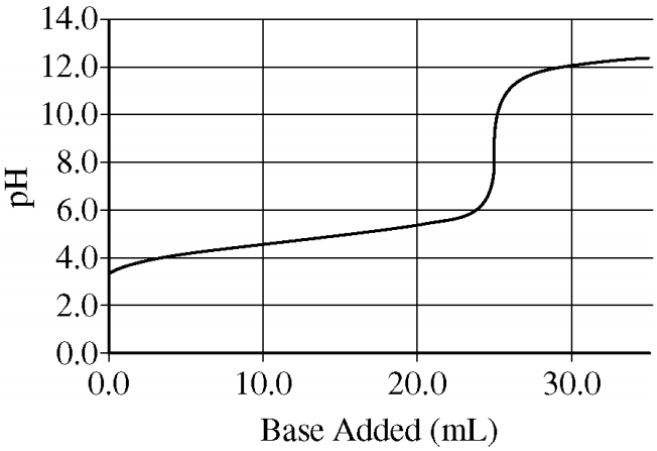
The rate law for the reaction shown is $rate = k[NO]^2[H_2]$. What is the overall order of the reaction?

unit5mcq

(A) First order
(B) Second order
(C) Third order ✓
(D) Fourth order

A 0.35 g sample of Li(*s*) is placed in an Erlenmeyer flask containing 100 mL of water at 25°C. A balloon is placed over the mouth of the flask to collect the hydrogen gas that is generated.

After all of the Li(*s*) has reacted with $H_2O(l)$, the solution in the flask is added to a clean, dry buret and used to titrate an aqueous solution of a monoprotic acid. The pH curve for this titration is shown in the diagram below



34. Which of the following changes will most likely increase the rate of reaction between Li(s) and water?

(A) Using 125 mL of water instead of 100 mL
(B) Using a 0.25 g sample of Li(*s*) instead of a 0.35 g sample
(C) Using a 0.35 g sample of Li(*s*) cut into small pieces ✓
(D) Decreasing the water temperature before adding the Li(*s*)

---

35. The half-life of ^{55}Cr is about 2.0 hours. The delivery of a sample of this isotope from the reactor to a certain laboratory requires 12 hours. About what mass of such material should be shipped in order that 1.0 mg of ^{55}Cr is delivered to the laboratory?

(A) 130 mg
(B) 64 mg ✓
(C) 32 mg
(D) 11 mg
(E) 1.0 mg

---

unit5mcq

36.

Experiment	Initial [NO] (mol L^{-1})	Initial [O_2] (mol L^{-1})	Initial Rate of Formation of NO_2 (mol $L^{-1}s^{-1}$)
1	0.10	0.10	2.5×10^{-4}
2	0.20	0.10	5.0×10^{-4}
3	0.20	0.40	8.0×10^{-3}

The initial-rate data in the table above were obtained for the reaction represented below. What is the experimental rate law for the reaction?

$$2\ NO(g) + O_2(g) \rightarrow NO_2(g)$$

(A) Rate = $k[NO][O_2]$

(B) Rate = $k[NO][O_2]^2$ ✓

(C) Rate = $k[NO]^2[O_2]$

(D) Rate = $k[NO]^2[O_2]^2$

(E) Rate = $k[NO] / [O_2]$

37.

Step 1:	?	(*slow*)
Step 2:	$NO_3(g) + CO(g) \rightarrow NO_2(g) + CO_2(g)$	(*fast*)
Overall:	$NO_2(g) + CO(g) \rightarrow NO(g) + CO_2(g)$	

A two-step reaction mechanism is proposed for a gas-phase reaction, as represented above. Which of the following correctly identifies both the chemical equation for step 1 and the rate law for the overall reaction?

(A) The chemical equation for step 1 is $2\ NO_2(g) \rightarrow NO(g) + NO_3(g)$, and the rate law is $rate = [NO_2]^2$. ✓

(B) The chemical equation for step 1 is $NO_2(g) + CO(g) \rightarrow NO(g) + CO_2(g)$, and the rate law is $rate = [NO_2][CO]$.

(C) The chemical equation for step 1 is $NO_3(g) + 2\ CO(g) \rightarrow NO(g) + 2\ CO_2(g)$, and the rate law is $rate = [NO_2][CO]$.

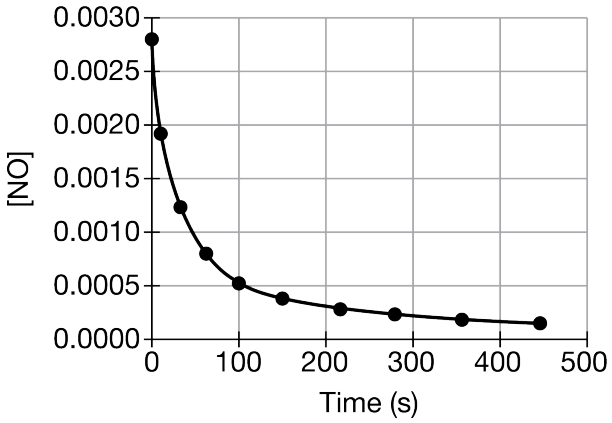
(D) The chemical equation for step 1 is $NO_3(g) + NO_2(g) + 2\ CO(g) \rightarrow NO(g) + NO_2(g) + 2\ CO_2(g)$, and the rate law is $rate = [NO_3][NO_2][CO]^2$.

unit5mcq

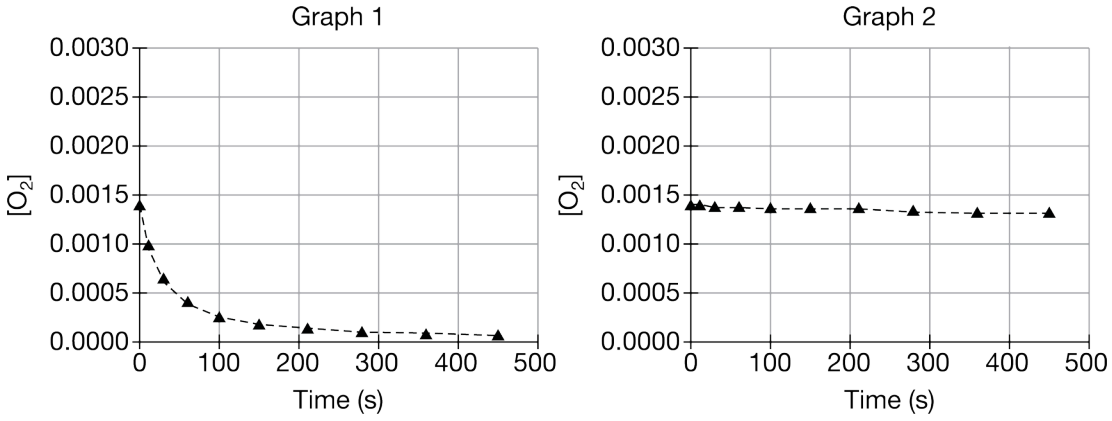
38. $2\ \mathrm{NO}(g) + \mathrm{O_2}(g) \rightarrow 2\ \mathrm{NO_2}(g)$

Reactant	Initial Concentration
NO	$0.0028\ M$
$\mathrm{O_2}$	$0.0014\ M$

The oxidation of $\mathrm{NO}(g)$ producing $\mathrm{NO_2}(g)$ is represented by the chemical equation shown above. The initial concentrations of NO and $\mathrm{O_2}$ are given in the table above. The changes in the concentration of $\mathrm{NO}(g)$ as a function of time are shown in the following graph.



Under the same conditions, one of the following graphs represents the changes in the concentration of $\mathrm{O_2}(g)$ over the same period of time.



Which graph best represents the changes in concentration of $\mathrm{O_2}(g)$, and why?

unit5mcq

(A) Graph 1, because the rate of O_2 consumption is half the rate at which NO is consumed; two molecules of NO react for each molecule of O_2 that reacts. ✓

(B) Graph 1, because O_2 molecules are consumed at a slower rate at the beginning of the reaction when there are not as many molecules of NO_2 produced.

(C) Graph 2, because there is a large excess of O_2 molecules and its concentration will not change drastically over time.

(D) Graph 2, because a collision between three molecules to form a product has a low probability and the concentration of O_2 will remain relatively constant.

39.

Experiment	$[X]_0$	$[Y]_0$	Initial Rate of Formation of Z (mol L^{-1} sec^{-1})
1	0.40	0.10	*R*
2	0.20	0.20	?

The table above shows the results from a rate study of the reaction X + Y → Z. Starting with known concentrations of X and Y in experiment 1, the rate of formation of Z was measured. If the reaction was first order with respect to X and second order with respect to Y, the initial rate of formation of Z in experiment 2 would be

(A) $\frac{R}{4}$

(B) $\frac{R}{2}$

(C) R

(D) $2R$ ✓

(E) $4R$

40. A $100.$ g sample of a substance decomposes so that 25 g of the original substance remains after $40.$ seconds. If the decomposition is first order, which of the following is the rate constant, k, of the decomposition reaction?

(A) $0.017\ s^{-1}$

(B) $0.035\ s^{-1}$ ✓

(C) $1.4\ s^{-1}$

(D) $20.\ s^{-1}$

41.

Trial	$[A_2]$	$[B]$	**Initial rate** (Ms^{-1})
1	0.10	0.50	2.5×10^{-4}
2	0.20	0.50	5.0×10^{-4}
3	0.30	0.05	5.0×10^{-5}
4	0.30	0.10	1.0×10^{-4}

An experiment was conducted to determine the rate law for the reaction $A_2(g) + B(g) \rightarrow A_2B(g)$. The table above shows the data collected. Based on the data in the table, which statement is correct?

unit5mcq

(A) Since the rate law can be expressed as $rate = k[\mathrm{A_2}]^2$, tripling the concentration of $\mathrm{A_2}$ will cause a 9-fold increase in the rate of the reaction.

(B) Since the rate law can be expressed as $rate = k[\mathrm{A_2}][\mathrm{B}]$, doubling the concentrations of $\mathrm{A_2}$ and B will quadruple the rate of the reaction. ✓

(C) Since the rate law can be expressed as $rate = k[\mathrm{A_2}]^2[\mathrm{B}]$, tripling the concentration of $\mathrm{A_2}$ while keeping the concentration of B constant will triple the rate of the reaction.

(D) Since the rate law can be expressed as $rate = k[\mathrm{A_2}][\mathrm{B}]^2$, doubling the concentration of B while keeping the concentration of $\mathrm{A_2}$ constant will double the rate of the reaction.

42. $2\ \mathrm{NO}(g) + 2\ \mathrm{H_2}(g) \rightarrow \mathrm{N_2}(g) + 2\ \mathrm{H_2O}(g)$

The rate law for the reaction represented is $rate = k[\mathrm{NO}]^2[\mathrm{H_2}]$. What happens to the rate of the reaction when the concentration of $\mathrm{NO}(g)$ is doubled and the concentration of $\mathrm{H_2}(g)$ is halved?

(A) The rate is halved.

(B) The rate is the same.

(C) The rate is doubled. ✓

(D) The rate is quadrupled.

43.

$$\text{Rate} = k[\text{M}][\text{N}]^2$$

The rate of a certain chemical reaction between substances M and N obeys the rate law above. The reaction is first studied with [M] and [N] each 1 x 10^{-3} molar. If a new experiment is conducted with [M] and [N] each 2 x 10^{-3} molar, the reaction rate will increase by a factor of

(A) 2

(B) 4

(C) 6

(D) 8 ✓

(E) 16

44.

Time (s)	$[\mathrm{NO_2}]$ $(\mathrm{mol/L})$	$\ln[\mathrm{NO_2}]$	$\frac{1}{[\mathrm{NO_2}]}$ $(\mathrm{L/mol})$
0	0.500	−0.693	2.00
100	0.364	−1.01	2.75
200	0.286	−1.25	3.50
300	0.235	−1.45	4.25

The data from a study of the decomposition of $\mathrm{NO_2}(g)$ to form $\mathrm{NO}(g)$ and $\mathrm{O_2}(g)$ are given in the table above. Which of the following rate laws is consistent with the data?

unit5mcq

(A) $\text{Rate} = k[\text{NO}_2]$

(B) $\text{Rate} = k[\text{NO}_2]^2$ ✓

(C) $\text{Rate} = k\frac{1}{[\text{NO}_2]}$

(D) $\text{Rate} = k\frac{1}{[\text{NO}_2]^2}$

45. $\text{NO}(g)$ and $\text{O}_2(g)$ react to form $\text{NO}_2(g)$. The rate law of the reaction is $rate = k[\text{NO}]^2\left[\text{O}_2\right]$. If the reaction occurs in a single elementary step that is a three-body molecular collision, then which of the following is the equation for the elementary step?

(A) $\text{N} + \text{O} + \text{O} \rightarrow \text{NO}_2$

(B) $2\,\text{NO} + \text{O}_2 \rightarrow 2\,\text{NO}_2$ ✓

(C) $\text{N} + \text{O}_2 \rightarrow \text{NO}_2$

(D) $\text{N}_2 + 2\,\text{O}_2 \rightarrow 2\,\text{NO}_2$

46. $2\,\text{NO}(g) + \text{Cl}_2(g) \rightarrow 2\,\text{NOCl}(g)$

Experiment	$[\text{NO}]$	$[\text{Cl}_2]$	**Initial Rate of NOCl Formation** $\left(M\,\text{s}^{-1}\right)$
1	0.0250	0.0510	4.55×10^{-5}
2	0.0250	0.1020	9.10×10^{-5}
3	0.0500	0.0510	1.82×10^{-4}

The initial rates of the reaction represented by the equation shown above were measured for different initial concentrations of $\text{NO}(g)$ and $\text{Cl}_2(g)$. Based on the data given in the table above, which of the following is the rate law expression for the reaction, and why?

(A) $\text{Rate} = k[\text{NO}]^2$, because the initial rate quadrupled when $[\text{NO}]$ was doubled but remained constant when $[\text{Cl}_2]$ was doubled.

(B) $\text{Rate} = k[\text{NO}][\text{Cl}_2]$, because the initial rate doubled when either $[\text{NO}]$ or $[\text{Cl}_2]$ was doubled.

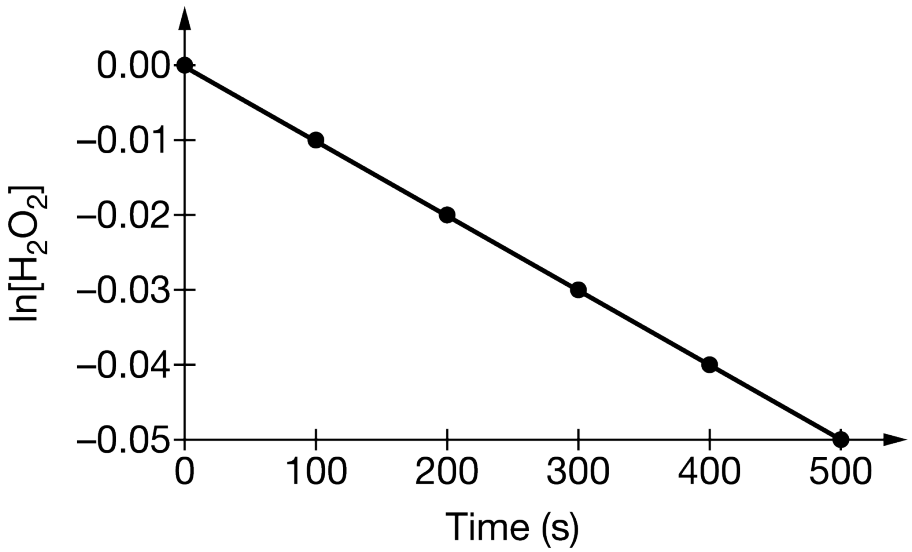
(C) $\text{Rate} = k[\text{NO}][\text{Cl}_2]^2$, because the initial rate doubled when $[\text{NO}]$ was doubled and quadrupled when $[\text{Cl}_2]$ was doubled.

(D) $\text{Rate} = k[\text{NO}]^2[\text{Cl}_2]$, because the initial rate quadrupled when $[\text{NO}]$ was doubled and doubled when $[\text{Cl}_2]$ was doubled. ✓

unit5mcq

47. $H_2O_2 \rightarrow H_2O + \frac{1}{2}O_2$

Hydrogen peroxide decomposes to produce water and oxygen according to the equation above. An experimentally determined graph for the first-order decomposition of hydrogen peroxide is provided below.



Which of the following best identifies the rate constant k for the reaction based on the information in the plot of $\ln[H_2O_2]$ versus time (t) ?

(A) $k = \ln[H_2O_2]$ at $t = 0$ s

(B) $k = \ln[H_2O_2]$ at $t = 500$ s

(C) $k = -(\text{slope of plot})$ ✓

(D) $k = -\frac{1}{(\text{slope of plot})}$

48. $S_2O_8^{2-}(aq) + 3\,I^-(aq) \rightarrow 2\,SO_4^{2-}(aq) + I_3^-(aq)$

In aqueous solution, the reaction represented by the balanced equation shown above has the experimentally determined rate law: $\text{rate} = k\left[S_2O_8^{2-}\right]\left[I^-\right]$.

If the concentration of $\left[S_2O_8^{2-}\right]$ is doubled while keeping $\left[I^-\right]$ constant, which of the following experimental results is predicted based on the rate law, and why?

(A) The rate of reaction will remain the same, because k will decrease by half.

(B) The rate of reaction will double, because the rate is directly proportional to $[S_2O_8^{2-}]$. ✓

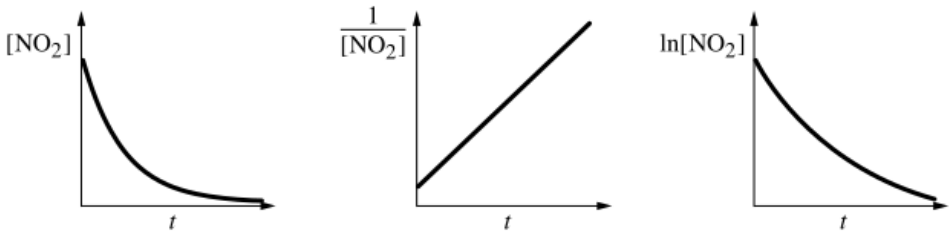
(C) The rate of reaction will increase by a factor of four, because two moles of SO_4^{2-} are produced for each mole of $S_2O_8^{2-}$ consumed.

(D) The rate of reaction will increase by a factor of four, because the reaction is second order overall.

unit5mcq

49. $\mathrm{NO_2}(g) + \mathrm{CO}(g) \rightarrow \mathrm{NO}(g) + \mathrm{CO_2}(g)$

A group of students studied the rate of the reaction represented above at a fixed temperature. During the reaction, the concentration of CO was in large excess. The graphs shown below were created using the data collected by the students. Based on the graphs, which of the following is the rate law?



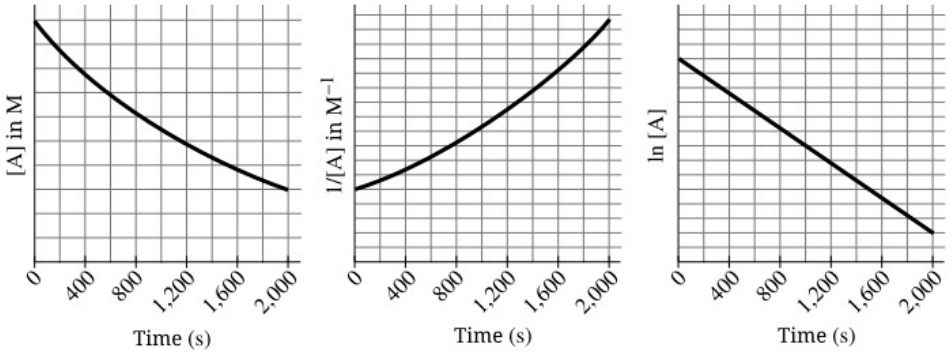
(A) $\text{Rate} = k$

(B) $\text{Rate} = [\mathrm{NO_2}]$

(C) $\text{Rate} = k[\mathrm{NO_2}]$

(D) $\text{Rate} = k[\mathrm{NO_2}]^2$ ✓

50.



A student recorded the rate of decomposition of compound A as a function of time at $45\,^\circ\mathrm{C}$. The experimental data are shown above. What is the rate law for the decomposition of A at this temperature?

(A) $\text{Rate} = k$

(B) $\text{Rate} = k[\mathrm{A}]$ ✓

(C) $\text{Rate} = k[\mathrm{A}]^2$

(D) The rate law cannot be determined without additional experimentation.

51. The rate law for the reaction of nitrogen dioxide and chlorine is found to be rate = $k\,[\mathrm{NO_2}]^2[\mathrm{Cl_2}]$. By what factor does the rate of the reaction change when the concentrations of both $\mathrm{NO_2}$ and $\mathrm{Cl_2}$ are doubled?

unit5mcq

(A) 2
(B) 3
(C) 4
(D) 6
(E) 8 ✓

52.

Experiment	*Initial [X] (mol L^{-1})*	*Initial [Y] (mol L^{-1})*	*Initial Rate of Formulation of Z (mol L^{-1} min^{-1})*
1	0.10	0.30	4.0×10^{-4}
2	0.20	0.60	1.6×10^{-3}
3	0.20	0.30	4.0×10^{-4}

The data in the table above were obtained for the reaction X + Y → Z. Which of the following is the rate law for the reaction?

(A) Rate = $k[X]^2$
(B) Rate = $k[Y]^2$ ✓
(C) Rate = $k[X][Y]$
(D) Rate = $k[X]^2[Y]$
(E) Rate = $k[X][Y]^2$

53.

Experiment	Initial [*X*] (mol L^{-1})	Initial [*Y*] (mol L^{-1})	Initial Rate of Formation of *Z* (mol L^{-1} s^{-1})
1	0.10	0.30	4.0×10^{-4}
2	0.20	0.60	1.6×10^{-3}
3	0.20	0.30	4.0×10^{-4}

The data in the table above were obtained for the reaction $X + Y \rightarrow Z$. Which of the following is the rate law for the reaction?

(A) Rate = $k[X]^2$
(B) Rate = $k[Y]^2$ ✓
(C) Rate = $k[X][Y]$
(D) Rate = $k[X]^2[Y]$
(E) Rate = $k[X][Y]^2$

unit5mcq

The following balanced equation represents the decomposition of hydrogen peroxide, H_2O_2.

$$2\ H_2O_2(aq)\ \rightarrow\ 2\ H_2O(l)\ +\ O_2(g)$$

A student carries out an experiment in which a sample of $H_2O_2(aq)$ decomposes in the presence of $MnO_2(s)$. The student measures the volume of $O_2(g)$ produced by the reaction at regular intervals while the temperature is held constant. The student's data are given in the following table.

Time (s)	Volume of $O_2(g)$ Collected (mL)
0	0
60	42
120	64
180	76
240	82
300	86
360	88
420	89

54. Based on the student's data, which of the following statements correctly describes how the rate of O_2 production changes during the experiment?

(A) The rate is constant at first and then increases over time.

(B) The rate decreases throughout the experiment. ✓

(C) The rate increases throughout the experiment.

(D) The rate is constant throughout the experiment.

55. A second experiment is done under identical conditions using a higher concentration of $H_2O_2(aq)$ in the presence of the same mass of $MnO_2(s)$. The volume of gas collected during the first 60 seconds will most likely be

(A) less than $42\ mL$

(B) equal to $42\ mL$

(C) greater than $42\ mL$ ✓

(D) impossible to predict without knowing the actual concentration of H_2O_2 used in the two experiments

56. The student observes that under the same conditions, the reaction occurs at a much slower rate when $MnO_2(s)$ is not used. Which of the following statements best explains this observation?

unit5mcq

(A) $MnO_2(s)$ changes the reaction from endothermic to exothermic.

(B) $MnO_2(s)$ shifts the equilibrium position of the reaction from reactant favored to product favored.

(C) $MnO_2(s)$ releases heat in order to start the reaction.

(D) $MnO_2(s)$ provides a reaction pathway with a lower activation energy. ✓

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57. $CaCO_3(s) + 2\ H^+(aq) \rightarrow Ca^{2+}(aq) + H_2O(l) + CO_2(g)$

Experiment	Initial Mass of $CaCO_3(s)$ (grams)	Initial Concentration of $H^+(aq)$ (M)	Initial Temperature of Reactants $(^\circ C)$
1	5.00	1.0	25.0
2	5.00	2.0	25.0

The initial rate of formation of $CO_2(g)$ from the chemical reaction represented by the equation above was studied in two separate experiments. The table above provides the experimental conditions used. If both experiments are carried out with finely powdered samples of the solid and $50.0\ mL$ of $HCl(aq)$, which experiment, if any, will have the faster initial rate of formation of $CO_2(g)$ and why?

(A) The rate of formation of $CO_2(g)$ will be the same because the mass of $CaCO_3(s)$ and the volume of $HCl(aq)$ used will be the same in both experiments.

(B) The rate of formation of $CO_2(g)$ will be the same because the surface area of the solid and the average kinetic energy of the particles will be the same in both experiments.

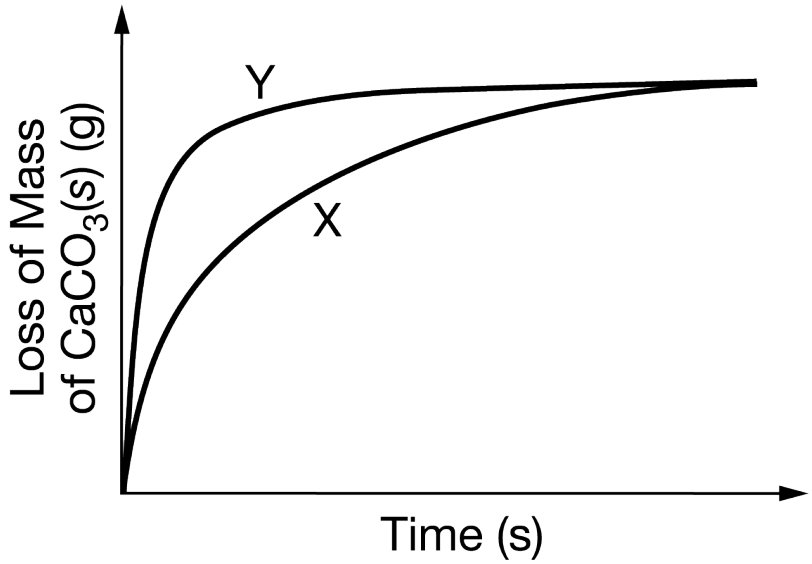
(C) $CO_2(g)$ will be formed at a faster rate in experiment 2 because more H^+ particles can react per unit time. ✓

(D) $CO_2(g)$ will be formed at a faster rate in experiment 1 because the proportion of $CaCO_3(s)$ particles to H^+ particles will be greater.

unit5mcq

58. $\mathrm{HCl}(aq) + \mathrm{CaCO_3}(s) \rightarrow \mathrm{CaCl_2}(aq) + \mathrm{CO_2}(g) + \mathrm{H_2O}(l)$

The reaction between $\mathrm{HCl}(aq)$ and $\mathrm{CaCO_3}(s)$ is represented by the equation above. Two separate trials were carried out using $\mathrm{CaCO_3}(s)$ samples of the same mass, but one sample was a single piece of $\mathrm{CaCO_3}(s)$, and one sample was composed of small pieces of $\mathrm{CaCO_3}(s)$. The loss of mass of $\mathrm{CaCO_3}(s)$ as a function of time for both trials is shown in the graph below.



Which of the curves, X or Y, represents the reaction with small pieces of $\mathrm{CaCO_3}(s)$, and why?

(A) Curve X, because it shows that the reaction proceeded at a uniform rate.

(B) Curve X, because it takes a shorter time for the reaction to go to completion due to the larger surface area of $\mathrm{CaCO_3}(s)$.

(C) Curve Y, because it shows that the reaction proceeded at a nonuniform rate.

(D) Curve Y, because it takes a shorter time for the reaction to go to completion due to the larger surface area of $\mathrm{CaCO_3}(s)$. ✓

59. When the chemical reaction $2\,\mathrm{NO}(g) + \mathrm{O_2}(g) \rightarrow 2\,\mathrm{NO_2}(g)$ is carried out under certain conditions, the rate of disappearance of $\mathrm{NO}(g)$ is $5.0 \times 10^{-5}\ M\mathrm{s}^{-1}$. What is the rate of disappearance of $\mathrm{O_2}(g)$ under the same conditions?

(A) Because two molecules of NO are consumed per molecule of $\mathrm{O_2}$, the rate of disappearance of $\mathrm{O_2}(g)$ is $2.5 \times 10^{-5}\ M\mathrm{s}^{-1}$. ✓

(B) Because two molecules of NO are consumed per molecule of $\mathrm{O_2}$, the rate of disappearance of $\mathrm{O_2}(g)$ is $1.0 \times 10^{-4}\ M\mathrm{s}^{-1}$.

(C) Because the rate depends on $[\mathrm{NO}]^2$ and $[\mathrm{O_2}]$, the rate of disappearance of $\mathrm{O_2}(g)$ is $2.5 \times 10^{-9}\ M\mathrm{s}^{-1}$.

(D) Because the $\mathrm{O_2}(g)$ is under the same experimental conditions as NO, it is consumed at the same rate of $5.0 \times 10^{-5}\ M\mathrm{s}^{-1}$.

unit5mcq

60. Relatively slow rates of chemical reaction are associated with which of the following?

(A) The presence of a catalyst

(B) High temperature

(C) High concentration of reactants

(D) Strong bonds in reactant molecules ✓

(E) Low activation energy

61.

$$X(g) + 2\ Y(g) \rightarrow XY_2(g)$$

In order to determine the order of the reaction represented above, the initial rate of formation of XY_2 is measured using different initial values of [X] and [Y]. The results of the experiment are shown in the table below.

Trial	[X]	[Y]	Initial Rate of Formation of XY_2 ($M\ s^{-1}$)
1	0.50	0.50	8.0×10^{-3}
2	1.00	0.50	3.2×10^{-2}
3	1.00	1.00	6.4×10^{-2}

In trial 2 which of the reactants would be consumed more rapidly, and why?

(A) X, because it has a higher molar concentration.

(B) X, because the reaction is second order with respect to X.

(C) Y, because the reaction is second order with respect to Y.

(D) Y, because the rate of disappearance will be double that of X. ✓

62.

Time (days)	0	1	2	3	4	5	6	7	. . .	10	. . .	20
% Reactant remaining	100	79	63	50	40	31	25	20		10		1

A reaction was observed for 20 days and the percentage of the reactant remaining after each day was recorded in the table above. Which of the following best describes the order and the half-life of the reaction?

unit5mcq

(A)

Reaction Order	Half-life (days)
First	3

✓

(B)

Reaction Order	Half-life (days)
First	10

(C)

Reaction Order	Half-life (days)
Second	3

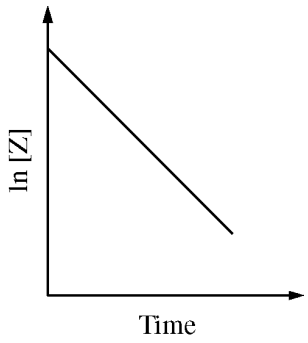
(D)

Reaction Order	Half-life (days)
Second	3

(E)

Reaction Order	Half-life (days)
Second	10

63.



Consider the reaction represented by the equation $2\ X + 2\ Z \rightarrow X_2Z_2$. During a reaction in which a large excess of reactant X was present, the concentration of reactant Z was monitored over time. A plot of the natural logarithm of the concentration of Z versus time is shown in the figure above. The order of the reaction with respect to reactant Z is

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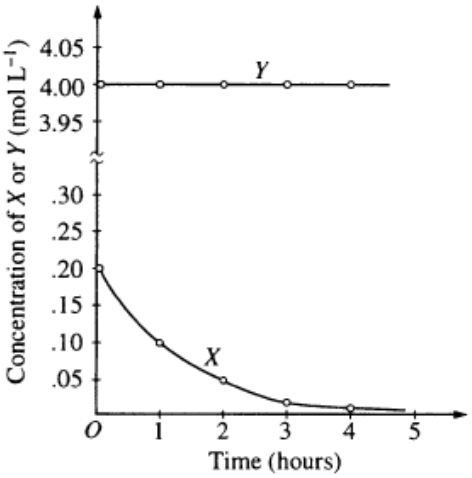
(A) zero order

(B) first order ✓

(C) second order

(D) third order

64.



The graph above shows the results of a study of the reaction of *X* with a large excess of *Y* to yield *Z*. The concentrations of *X* and *Y* were measured over a period of time. According to the results, which of the following can be concluded about the rate law for the reaction under the conditions studied?

(A) It is zero order in [*X*].

(B) It is first order in [*X*]. ✓

(C) It is second order in [*X*].

(D) It is first order in [*Y*].

(E) The overall order of the reaction is 2.

65. $NO_2\,Cl(g) + Cl\,(g) \rightarrow NO_2(g) + Cl_2(g)$

The single-step reaction represented above takes place in a sealed container. Which of the following will reduce the value of k, the rate constant for the reaction?

(A) Decreasing the temperature ✓

(B) Increasing the temperature

(C) Decreasing the size of the sealed container

(D) Increasing the size of the sealed container

unit5mcq

66.

$$X + 2\ Y \rightarrow Z + 3\ Q$$

For the reaction represented above, the initial rate of decrease in [X] was 2.8×10^{-3} mol L^{-1} s^{-1}. What was the initial rate of decrease in [Y] ?

(A) 7.0×10^{-4} mol L^{-1} s^{-1}

(B) 1.4×10^{-3} mol L^{-1} s^{-1}

(C) 2.8×10^{-3} mol L^{-1} s^{-1}

(D) 5.6×10^{-3} mol L^{-1} s^{-1} ✓

(E) 1.1×10^{-2} mol L^{-1} s^{-1}

67. 2 A*(g)* + B*(g)* → 2 C*(g)*

When the concentration of substance B in the reaction above is doubled, all other factors being held constant, it is found that the rate of the reaction remains unchanged. The most probable explanation for this observation is that

(A) the order of the reaction with respect to substance B is 1

(B) substance B is not involved in any of the steps in the mechanism of the reaction

(C) substance B is not involved in the rate–determined step of the mechanism, but is involved in subsequent steps ✓

(D) substance B is probably a catalyst, and as such, its effect on the rate of the reaction does not depend on its concentration

(E) the reactant with the smallest coefficient in the balanced equation generally has little or no effect on the rate of the reaction

68.

$$X_2 + Y_2 \rightarrow X_2Y_2$$

$$\text{rate} = k[X_2]$$

A reaction and its experimentally determined rate law are represented above. A chemist proposes two different possible mechanisms for the reaction, which are given below.

Mechanism 1	Mechanism 2
$X_2 \rightarrow 2\ X$ (*slow*)	$X_2 \rightarrow 2\ X$ (*slow*)
$X + Y_2 \rightarrow XY_2$ (*fast*)	$X + Y_2 \rightarrow XY + Y$ (*fast*)
$X + XY_2 \rightarrow X_2Y_2$ (*fast*)	$X + XY \rightarrow X_2Y$ (*fast*)
	$X_2Y + Y \rightarrow X_2Y_2$ (*fast*)

Based on the information above, which of the following is true?

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(A) Only mechanism 1 is consistent with the rate law.

(B) Only mechanism 2 is consistent with the rate law.

(C) Both mechanism 1 and mechanism 2 are consistent with the rate law. ✓

(D) Neither mechanism 1 nor mechanism 2 is consistent with the rate law.

69.

$$2\,NO_2(g) + F_2(g) \rightarrow 2\,NO_2F(g)$$

The rate law for the reaction represented by the equation above is rate = $k[NO_2][F_2]$. Which of the following could be the first elementary step of a two-step mechanism for the reaction if the first step is slow and the second step is fast?

(A) $F_2(g) \rightarrow 2\,F(g)$

(B) $NO_2(g) + F_2(g) \rightarrow NO_2F(g) + F(g)$ ✓

(C) $NO_2(g) + F(g) \rightarrow NO_2F(g)$

(D) $2\,NO_2(g) + F_2(g) \rightarrow 2\,NO_2F(g)$

70.

$$2NO(g) + O_2(g) \rightarrow N_2O_2(g)$$

A possible mechanism for the overall reaction represented above is the following.

1. $NO(g) + NO(g) \rightarrow N_2O_2(g)$ *slow*
2. $N_2O_2(g) + O_2(g) \rightarrow 2NO_2(g)$ *fast*

Which of the following rate expressions agrees best with this possible mechanism?

(A) Rate = $k[NO]^2$ ✓

(B) Rate = $k\frac{[NO]}{[O_2]}$

(C) Rate = $k\frac{[NO]^2}{[O_2]}$

(D) Rate = $k[NO]^2\,[O_2]$

(E) Rate = $k[N_2O_2]\,[O_2]$

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$4\ \mathrm{FeS}(s)\ +\ 7\ \mathrm{O_2}(g)\ \rightarrow\ 2\ \mathrm{Fe_2O_3}(s)\ +\ 4\ \mathrm{SO_2}(g)$

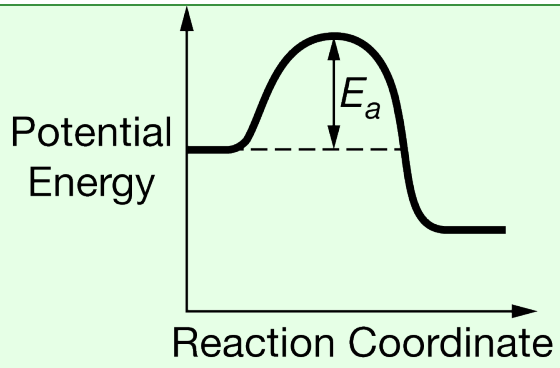
Due to the presence of $\mathrm{FeS}(s)$ as an impurity, the combustion of some types of coal results in the formation of $\mathrm{SO_2}(g)$, as represented by the equation above. Also, $\mathrm{SO_2}(g)$ can react with $\mathrm{O_3}(g)$ to form $\mathrm{SO_3}(g)$, as represented by the equation below.

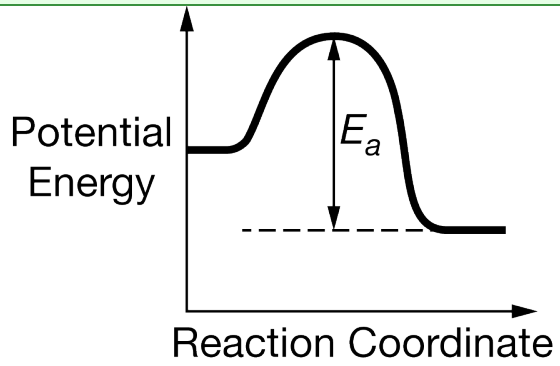
$\mathrm{SO_2}(g)\ +\ \mathrm{O_3}(g)\ \rightleftarrows\ \mathrm{SO_3}(g)\ +\ \mathrm{O_2}(g)$

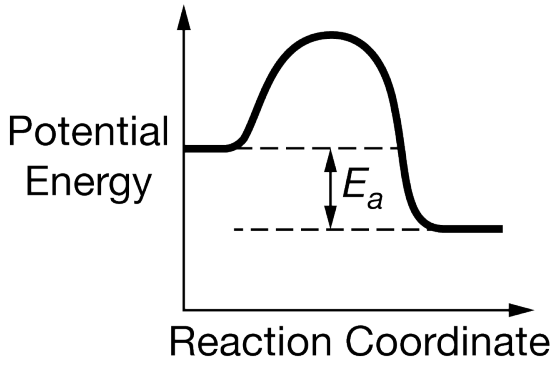
$\Delta H^\circ_{298} = -242\ \mathrm{kJ/mol}_{rxn}\ ;\ \Delta S^\circ_{298} = -25\ \mathrm{J/(K \cdot mol}_{rxn})$

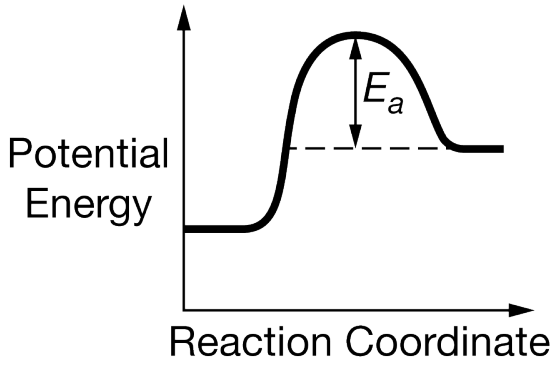
71. Which of the following diagrams shows the correctly labeled activation energy, E_a, for the reaction between $\mathrm{SO_2}(g)$ and $\mathrm{O_3}(g)$?

unit5mcq

(A) 

(B) 

(C) 

(D) 

---

72. $Cl^-(aq) + ClO^-(aq) + 2\ H^+(aq) \rightarrow Cl_2(g) + H_2O(l)$

What effect will increasing $[H^+]$ at constant temperature have on the reaction represented above?

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unit5mcq

(A) The activation energy of the reaction will increase.

(B) The activation energy of the reaction will decrease.

(C) The frequency of collisions between $H^+(aq)$ ions and $ClO^-(aq)$ ions will increase. ✓

(D) The value of the rate constant will increase.

When free $Cl(g)$ atoms encounter $O_3(g)$ molecules in the upper atmosphere, the following reaction mechanism is proposed to occur.

$$Cl(g) + O_3(g) \rightarrow ClO(g) + O_2(g) \qquad \textit{slow step}$$

$$ClO(g) + O_3(g) \rightarrow Cl(g) + 2\,O_2(g) \qquad \textit{fast step}$$

$$2\,O_3(g) \rightarrow 3\,O_2(g) \qquad \textit{overall reaction} \qquad \Delta H = -285\ \text{kJ/mol}_{rxn}$$

73. Which of the following rate laws for the overall reaction corresponds to the proposed mechanism?

(A) Rate = $k[O_3]^2$

(B) Rate = $k[Cl][O_3]$ ✓

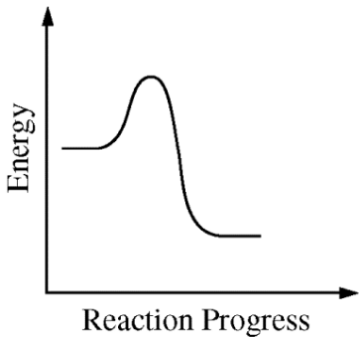
(C) Rate = $k[ClO][O_3]^2$

(D) Rate = $k[O_2]^3 / [\,O_3\,]^2$

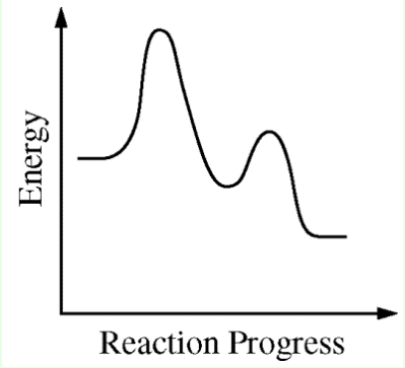
74. Which of the following reaction energy profiles best corresponds to the proposed mechanism?

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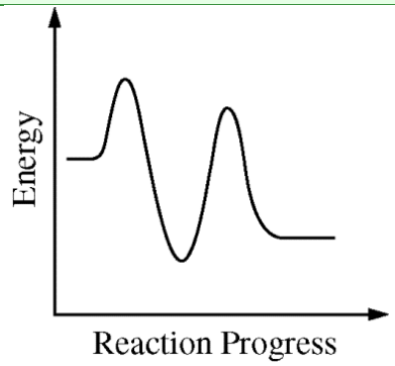
(A)



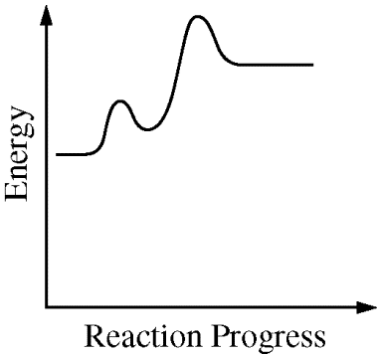
(B)



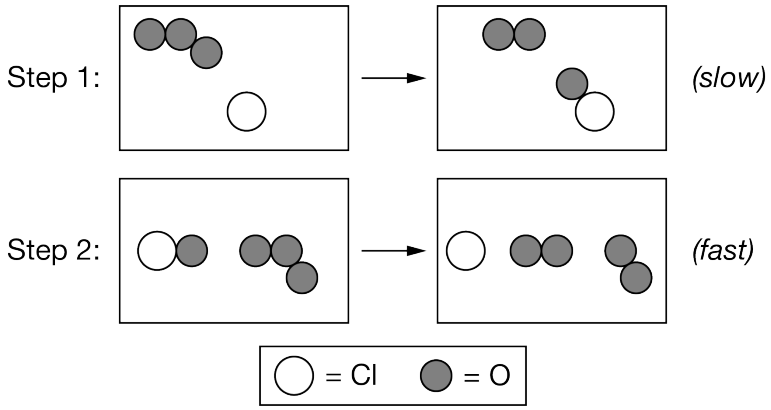
(C)



(D)

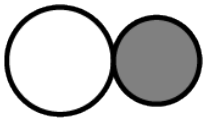


unit5mcq



The particle models shown above represent a proposed two-step mechanism for the destruction of ozone (O_3) in the upper atmosphere.

75.



Based on the proposed mechanism, which of the following best describes the concentration of the species represented above as the reaction occurs?

(A) The concentration is larger than that of reactant or product.

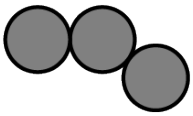
(B) The concentration is very low for the duration of the reaction. ✓

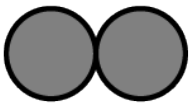
(C) The concentration increases steadily as the reaction proceeds.

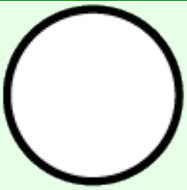
(D) The concentration fluctuates between high and low during the reaction.

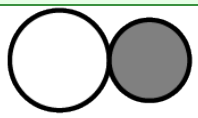
76. Based on the models, which of the following represents a species that acts as a catalyst for the reaction?

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(A) 

(B) 

(C)  ✓

(D) 

78. Based on the proposed mechanism, which of the following is the rate-law expression for the destruction of $\mathrm{O_3}$?

(A) $\text{Rate} = k[\mathrm{O_3}]^2$

(B) $\text{Rate} = k[\mathrm{Cl}]^2$

(C) $\text{Rate} = k[\mathrm{ClO}][\mathrm{O_3}]$

(D) $\text{Rate} = k[\mathrm{O_3}][\mathrm{Cl}]$ ✓

---

77. Based on the proposed mechanism, what is the balanced chemical equation for the overall reaction?

(A) $\mathrm{O_3}(g) + \mathrm{Cl}(g) \rightarrow \mathrm{O_2}(g) + \mathrm{ClO}(g)$

(B) $2\ \mathrm{O_3}(g) + 2\ \mathrm{Cl}(g) \rightarrow 2\ \mathrm{O_2}(g) + 2\ \mathrm{ClO}(g)$

(C) $\mathrm{O_3}(g) + \mathrm{Cl}(g) + \mathrm{ClO}(g) \rightarrow 2\ \mathrm{O_2}(g) + \mathrm{Cl_2}(g)$

(D) $2\ \mathrm{O_3}(g) \rightarrow 3\ \mathrm{O_2}(g)$ ✓

79. The gas-phase reaction $\mathrm{A_2}(g) + \mathrm{B_2}(g) \rightarrow 2\ \mathrm{AB}(g)$ is assumed to occur in a single step. Two experiments were done at the same temperature inside rigid containers. The initial partial pressures of $\mathrm{A_2}$ and $\mathrm{B_2}$ used in experiment 1 were twice the initial pressures used in experiment 2. Which statement provides the best comparison of the initial rate of formation of AB in experiments 1 and 2 ?

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(A) The initial rate of formation of AB is the same in both experiments because they were done at the same temperature and the frequency and energy of the collisions between $\mathrm{A_2}$ and $\mathrm{B_2}$ would have been about the same.

(B) The initial rate of formation of AB is slower in experiment 1 than in with experiment 2 because at the same temperature, a higher pressure would reduce the volume available for $\mathrm{A_2}$ and $\mathrm{B_2}$ molecules to achieve the proper orientation for a successful collision.

(C) The initial rate of formation of AB is faster in experiment 1 than in experiment 2 because at a higher pressure the collisions between $\mathrm{A_2}$ and $\mathrm{B_2}$ molecules would have been more frequent, increasing the probability of a successful collision. ✓

(D) The initial rate of formation of AB is faster in experiment 1 than in experiment 2 because at a higher pressure a larger fraction of the $\mathrm{A_2}$ and $\mathrm{B_2}$ molecules would have the minimum energy required to overcome the activation energy barrier.

80. $2\ \mathrm{N_2O}(g) \rightarrow 2\ \mathrm{N_2}(g) + \mathrm{O_2}(g)$

The rate law for the reaction represented above is experimentally determined to be $\mathrm{rate} = k[\mathrm{N_2O}]$. Which of the following best explains why increasing the initial concentration of $\mathrm{N_2O}(g)$ at a constant temperature increases the reaction rate?

(A) Molecular collisions become more frequent. ✓

(B) There is a higher probability of effective molecular collisions.

(C) The rate constant for the reaction increases.

(D) The reaction mechanism changes to one that has a lower activation energy.

81. Which of the following best helps explain why an increase in temperature increases the rate of a chemical reaction?

(A) At higher temperatures, reactions have a lower activation energy.

(B) At higher temperatures, reactions have a higher activation energy.

(C) At higher temperatures, every collision results in the formation of product.

(D) At higher temperatures, high-energy collisions happen more frequently. ✓

82. Factors that affect the rate of a chemical reaction include which of the following?

I. Frequency of collisions of reactant particles

II. Kinetic energy of collisions of reactant particles

III. Orientation of reactant particles during collisions

(A) II only

(B) I and II only

(C) I and III only

(D) II and III only

(E) I, II, and III ✓

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83. Step 1: $NO(g) + O_3(g) \rightarrow NO_2(g) + O_2(g)$

Step 2: $NO_2(g) + O(g) \rightarrow NO(g) + O_2(g)$

A reaction mechanism for the destruction of ozone, $O_3(g)$, is represented above. In the overall reaction, $NO(g)$ is best described as

(A) an inhibitor

(B) a catalyst ✓

(C) a reactant

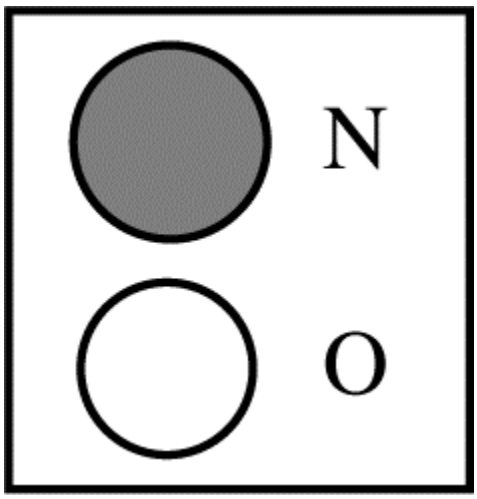
(D) an intermediate

(E) a product

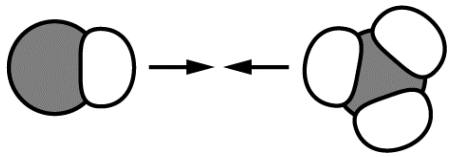
84. $NO(g) + NO_3(g) \rightarrow 2\ NO_2(g)$

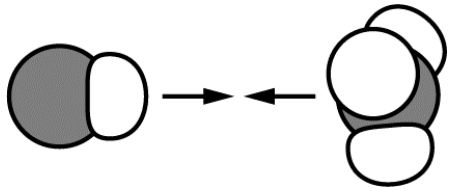
The reaction between $NO(g)$ and $NO_3(g)$ is represented by the equation above. Which of the

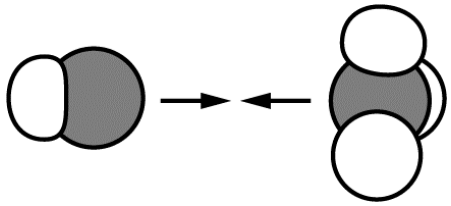
following orientations of collision between $NO(g)$ and $NO_3(g)$ is most likely to be effective?

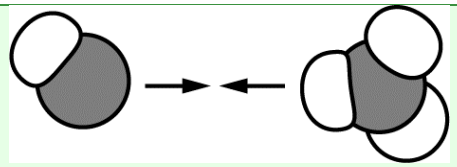


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(A) 

(B) 

(C) 

(D)  ✓

85. $NO2(g) + CO(g) \rightarrow NO(g) + CO2(g)$

The reaction between NO2(*g*) and CO(*g*) is represented above. The elementary steps of a proposed reaction mechanism are represented below.

Step 1: $2\ NO_2(g) \rightarrow NO(g) + NO_3(g)$ (*slow*)

Step 2: $NO_3(g) + CO(g) \rightarrow NO_2(g) + CO_2(g)$ (*fast*)

Which of the following is the rate law for the overall reaction that is consistent with the proposed mechanism?

(A) Rate = $k\ [NO_2][CO]$

(B) Rate = $k\ [NO_2]^2$ ✓

(C) Rate = $k\ [NO_3][CO]$

(D) Rate = $k\ [NO_2][NO_3][CO]$

86. $NO(g) + NO_3(g) \rightarrow 2\ NO_2(g)$

rate = $k[NO][NO_3]$

The reaction represented above occurs in a single step that involves the collision between a particle of NO and a particle of NO_3. A scientist correctly calculates the rate of collisions between NO and NO_3 that have sufficient energy to overcome the activation energy. The observed reaction rate is only a small fraction of the calculated collision rate. Which of the following best explains the discrepancy?

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(A) The energy of collisions between two reactant particles is frequently absorbed by collision with a third particle.

(B) The two reactant particles must collide with a particular orientation in order to react. ✓

(C) The activation energy for a reaction is dependent on the concentrations of the reactant particles.

(D) The activation energy for a reaction is dependent on the temperature.

87. $2\ \mathrm{NO}(g) + \mathrm{Br_2}(g) \rightarrow 2\ \mathrm{NOBr}(g)$

The two mechanisms in the table below have been proposed for the reaction represented above.

Mechanism 1		Mechanism 2	
Step 1: $\mathrm{NO}(g) + \mathrm{NO}(g) \rightarrow \mathrm{N_2O_2}(g)$	*slow*	Step 1: $\mathrm{NO}(g) + \mathrm{Br_2}(g) \rightarrow \mathrm{NOBr_2}(g)$	s
Step 2: $\mathrm{N_2O_2}(g) + \mathrm{Br_2}(g) \rightarrow 2\ \mathrm{NOBr}(g)$	*fast*	Step 2: $\mathrm{NOBr_2}(g) + \mathrm{NO}(g) \rightarrow 2\ \mathrm{NOBr}(g)$	f

Which of the following observations would support mechanism 1 but not mechanism 2 ?

(A) The reaction rate is independent of $[\mathrm{NO}]$.

(B) The reaction rate is independent of $[\mathrm{Br_2}]$. ✓

(C) The reaction is exothermic.

(D) The reaction is second order overall.

88. Step 1: $Ce^{4+} + Mn^{2+} \rightarrow Ce^{3+} + Mn^{3+}$

Step 2: $Ce^{4+} + Mn^{3+} \rightarrow Ce^{3+} + Mn^{4+}$

Step 3: $Mn^{4+} + Tl^{+} \rightarrow Tl^{3+} + Mn^{2+}$

The proposed steps for a catalyzed reaction between Ce^{4+} and Tl^{+} are represented above. The products of the overall catalyzed reaction are

(A) Ce^{4+} and Tl^{+}

(B) Ce^{3+} and Tl^{3+} ✓

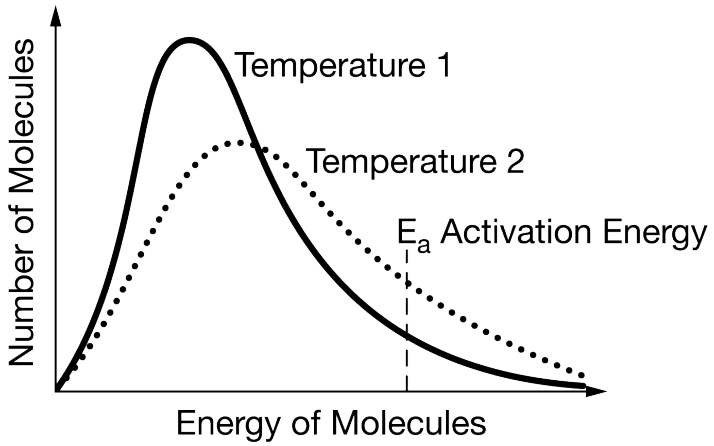
(C) Ce^{3+} and Mn^{3+}

(D) Ce^{3+} and Mn^{4+}

(E) Tl^{3+} and Mn^{2+}

unit5mcq

89.



The proposed rate-determining step for a reaction is $2\,NO_2(g) \rightarrow NO_3(g) + NO(g)$. The graph above shows the distribution of energies for $NO_2(g)$ molecules at two temperatures. Based on the graph, which of the following statements best explains why the rates of disappearance of $NO_2(g)$ are different at temperature 2 and temperature 1 ?

(A) $NO_2(g)$ is consumed at a faster rate at temperature 2 because more molecules possess energies at or above the minimum energy required for a collision to lead to a reaction compared to temperature 1. ✓

(B) $NO_2(g)$ is consumed at a faster rate at temperature 2 because the molecules have a wider range of energies allowing for a better orientation during a collision compared to temperature 1.

(C) Fewer $NO_2(g)$ molecules have a relatively high energy at temperature 1, which favors collisions between molecules rather than between the molecules and the container, leading to a faster rate of disappearance compared to temperature 2.

(D) More $NO_2(g)$ molecules have a relatively low energy at temperature 1, which increases the number of effective collisions taking place and the rate of disappearance compared to temperature 2.

90. $2\,HBr(g) + O_2(g) \rightarrow H_2O_2(g) + Br_2(g)$

Based on a kinetics study of the reaction represented by the equation above, the following mechanism for the reaction is proposed.

Step 1:	$HBr(g) + O_2(g) \rightarrow HO_2Br(g)$	(slow)
Step 2:	$HO_2Br(g) + HBr(g) \rightarrow 2\,HOBr(g)$	(fast)
Step 3:	$2\,HOBr(g) \rightarrow H_2O_2(g) + Br_2(g)$	(fast)

Which of following rate laws is consistent with the proposed mechanism?

unit5mcq

(A) $\text{Rate} = k[\text{HBr}]^2\left[\text{O}_2\right]$

(B) $\text{Rate} = k[\text{HBr}][\text{O}_2]$ ✓

(C) $\text{Rate} = k[\text{HO}_2\text{Br}][\text{HBr}]$

(D) $\text{Rate} = k\left[\text{HOBr}\right]^2$

91. $2\,\text{NO}_2 \rightarrow \text{NO}_3 + \text{NO} \quad (slow)$

$\text{NO}_3 \rightarrow \text{NO} + \text{O}_2 \quad (fast)$

A proposed two-step mechanism for the reaction $2\,\text{NO}_2 \rightarrow 2\,\text{NO} + \text{O}_2$ is represented. Which of the following rate laws is consistent with the mechanism?

(A) $\text{rate} = k[\text{NO}_3]$

(B) $\text{rate} = k[\text{NO}_2]^2$ ✓

(C) $\text{rate} = k[\text{NO}_2]^2[\text{NO}_3]$

(D) $\text{rate} = k\frac{[\text{NO}]^2[\text{O}_2]}{[\text{NO}_2]^2}$

92. Which of the following best describes the role of the spark from the spark plug in an automobile engine?

(A) The spark decreases the energy of activation for the slow step.

(B) The spark increases the concentration of the volatile reactant.

(C) The spark supplies some of the energy of activation for the combustion reaction. ✓

(D) The spark provides a more favorable activated complex for the combustion reaction.

(E) The spark provides the heat of vaporization for the volatile hydrocarbon.

unit5mcq

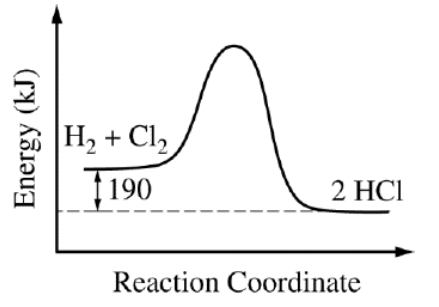
93.

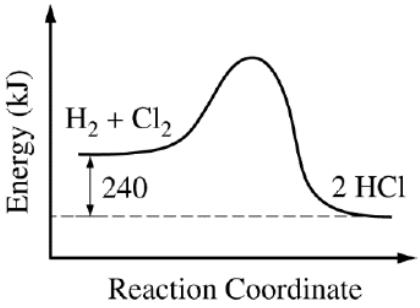
Bond	Bond Energy (kJ/mol)
H–H	430
Cl–Cl	240
H–Cl	430

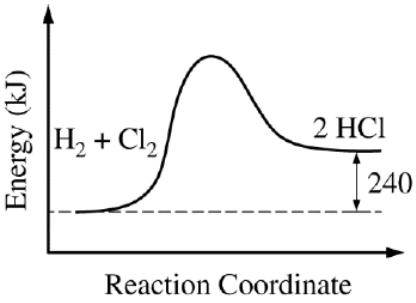
Based on the bond energies shown in the table above, which of the following diagrams best represents the change in energy as the reaction represented below proceeds?

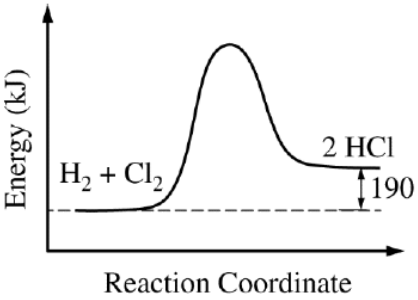
$$H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$$

unit5mcq

(A) 

(B) 

(C) 

(D) 

94. Which of the following statements best explains why an increase in temperature of 5-10 Celsius degrees can substantially increase the rate of a chemical reaction?

(A) The activation energy for the reaction is lowered.

(B) The number of effective collisions between reactant particles is increased. ✓

(C) The rate of the reverse reaction is increased.

(D) ΔH for the reaction is lowered.

(E) ΔG for the reaction becomes more positive.

unit5mcq

95. $2\ NO(g) + O_2(g) \rightarrow 2\ NO_2(g)$

Consider the following mechanism for the reaction represented above.

Step 1: $2\ NO \rightleftarrows N_2O_2$ (*fast reversible*)

Step 2: $N_2O_2 + O_2 \rightarrow 2\ NO_2$ (*slow*)

Which of the following statements is true?

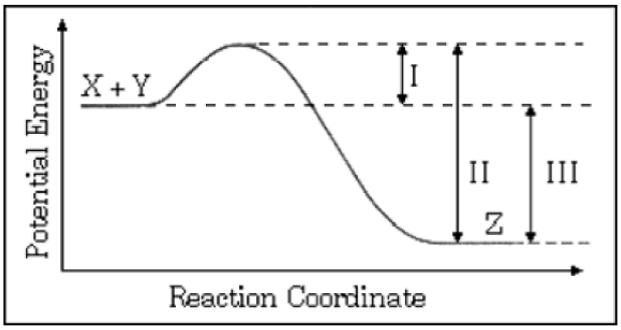
(A) Step 1 represents a unimolecular reaction.

(B) Increasing the concentration of NO will decrease the overall rate of the reaction.

(C) Raising the temperature will have no effect on the numerical value of the rate constant.

(D) The rate law that is consistent with the mechanism is *rate* = $k[NO]^2[O_2]$. ✓

96.



The energy diagram for the reaction X + Y → Z is shown above. The addition of a catalyst to this reaction would cause a change in which of the indicated energy differences?

(A) I only

(B) II only

(C) III only

(D) I and II only ✓

(E) I, II, and III

97. Step 1: $O_3(g) + Cl(g) \rightarrow O_2(g) + ClO(g)$Step 2: $ClO(g) + O(g) \rightarrow O_2(g) + Cl(g)$

A proposed mechanism for the destruction of $O_3(g)$ in the upper atmosphere is represented above. Which of the following is the catalyst in the proposed mechanism?

(A) $O(g)$

(B) $Cl(g)$ ✓

(C) $ClO(g)$

(D) $O_2(g)$

unit5mcq

98. $\mathrm{Br_2}(g) + 2\ \mathrm{NO}(g) \rightarrow 2\ \mathrm{NOBr}(g)$

The reaction represented by the equation above has the following proposed mechanism.

Step 1:	$\mathrm{NO}(g) + \mathrm{Br_2}(g) \rightleftharpoons \mathrm{NOBr_2}(g)$	(fast equilibrium)
Step 2:	$\mathrm{NOBr_2}(g) + \mathrm{NO}(g) \rightarrow 2\ \mathrm{NOBr}(g)$	(slow)

Based on the information, which of the following is the initial rate law for the reaction?

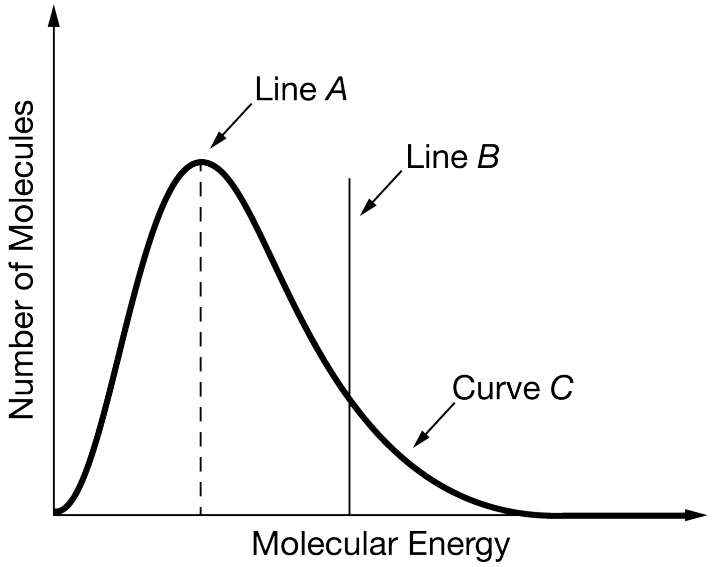
(A) $\text{Rate} = k[\mathrm{Br_2}][\mathrm{NO}]$

(B) $\text{Rate} = k\left[\mathrm{Br_2}\right][\mathrm{NO}]^2$ ✓

(C) $\text{Rate} = k[\mathrm{Br_2}]^2[\mathrm{NO}]$

(D) $\text{Rate} = k[\mathrm{NOBr_2}][\mathrm{NO}]$

99.



The energy distribution profile (Curve C) for the $\mathrm{Y_2}$ molecules is shown in the graph above for the reaction $\mathrm{X} + \mathrm{Y_2} \rightarrow \mathrm{XY_2}$ when it is done under certain experimental conditions. Line A represents the most probable energy of the $\mathrm{Y_2}$ molecules, and Line B represents the activation energy. Which of the following changes should be made to the graph to explain the increase in the rate of the reaction if the only change in experimental conditions is the addition of a catalyst?

(A) Curve C will broaden and Line B will move to the left, because more $\mathrm{Y_2}$ molecules will have an energy greater than the minimum energy needed to overcome the activation energy barrier.

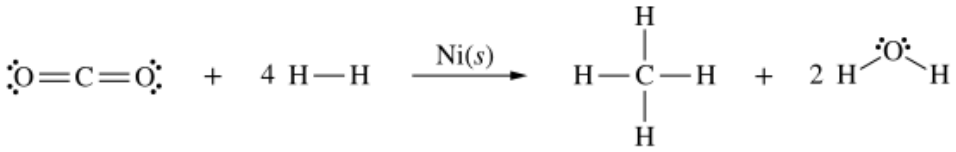
(B) Line B will move to the left because a larger fraction of the $\mathrm{Y_2}$ molecules will have the minimum energy to overcome the activation energy barrier. ✓

(C) Curve C and Line A will move to the right because the average energy of the $\mathrm{Y_2}$ molecules will increase.

(D) Line A will move to the right because the most probable energy of the $\mathrm{Y_2}$ molecules will increase.

unit5mcq

In the gas-phase reaction represented by the following equation, methane (CH_4) and water (H_2O) are produced from carbon dioxide (CO_2) and hydrogen (H_2) at high temperature and pressure.



100. The reaction is catalyzed by solid nickel. Which of the following describes the effect of the catalyst on the reaction?

(A) The presence of the catalyst results in an overall reaction that releases less energy.

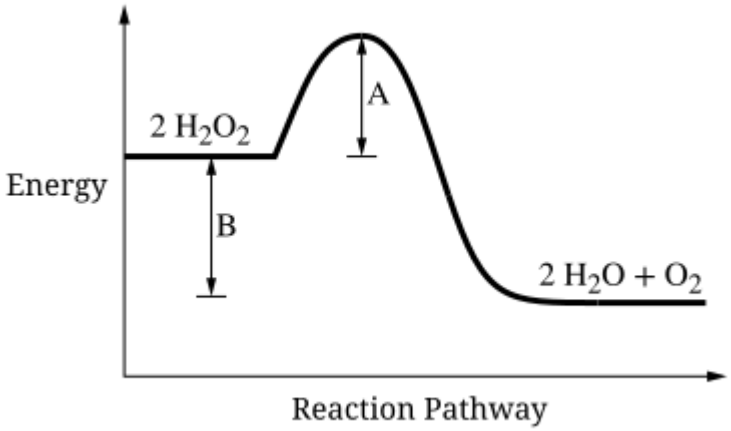
(B) The presence of the catalyst results in a reaction pathway that requires a lower activation energy. ✓

(C) The presence of the catalyst results in an overall reaction that requires more energy.

(D) The presence of the catalyst results in a reaction pathway that requires a greater activation energy.

unit5mcq

101.



The potential energy diagram for the uncatalyzed decomposition of $H_2O_2(l)$ is shown above. The arrows labeled A and B represent different amounts of energy. When $I^-(aq)$ is used as a catalyst, how do the magnitudes of A and B in the new reaction energy diagram compare to the uncatalyzed diagram shown above?

	Energy A	Energy B
A	Larger	Larger
B	Smaller	Smaller
C	Smaller	Unchanged
D	Unchanged	Smaller

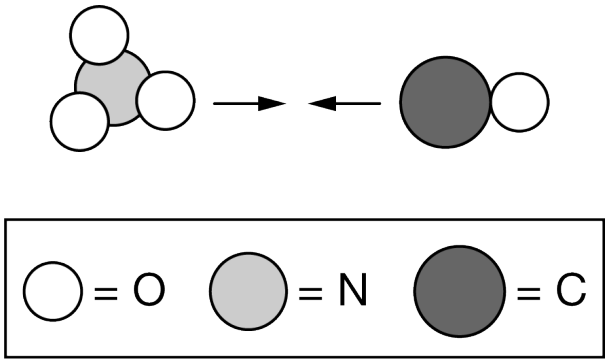
(A) A
(B) B
(C) C ✓
(D) D

102. To catalyze a biochemical reaction, an enzyme typically

(A) drives the reaction to completion by consuming byproducts of the reaction
(B) binds temporarily to reactant molecules to lower the activation energy of the reaction ✓
(C) dissociates into additional reactant molecules, thereby increasing the reaction rate
(D) decomposes and releases energy to increase the number of successful collisions between reactant molecules

unit5mcq

103.



The slowest step in a reaction mechanism requires the collision represented above to occur. Which of the following most likely indicates how the addition of a solid catalyst could increase the rate of the reaction?

(A) The catalyst could change the reaction from second order to third order.

(B) The catalyst could increase the particles' speed, thereby increasing the value of the rate constant, k.

(C) The catalyst could decrease the particles' speed, making it less likely that the particles will rebound without reacting when they collide.

(D) The catalyst could adsorb one of the particles, making a successful (reaction-producing) collision with the other particle more likely. ✓

104. $2\,H_2(g) + 2\,NO(g) \rightarrow N_2(g) + 2\,H_2O(g)$

The experimental rate law for the reaction represented above is $\text{rate} = k[H_2][NO]^2$. Which of the following proposed mechanisms is consistent with the rate law?

unit5mcq

(A)

Step 1: $2\,NO \rightarrow N_2O_2$ *fast*

Step 2: $H_2 + N_2O_2 \rightleftarrows N_2O + H_2O$ *fast*

Step 3: $H_2 + N_2O \rightarrow N_2 + H_2O$ *slow*

(B) ✓

Step 1: $2\,NO \rightleftarrows N_2O_2$ *fast*

Step 2: $H_2 + N_2O_2 \rightarrow N_2O + H_2O$ *slow*

Step 3: $H_2 + N_2O \rightarrow N_2 + H_2O$ *fast*

(C)

Step 1: $2\,NO \rightarrow N_2 + O_2$ *slow*

Step 2: $H_2 + O_2 \rightarrow H_2O_2$ *fast*

Step 3: $H_2 + H_2O_2 \rightarrow 2\,H_2O$ *fast*

(D)

Step 1: $2\,NO \rightarrow N_2 + O_2$ *fast*

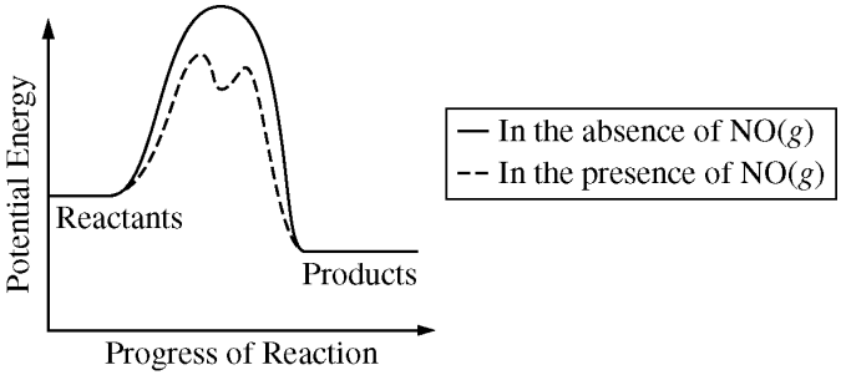
Step 2: $H_2 + O_2 \rightleftarrows H_2O_2$ *fast*

Step 3: $H_2 + H_2O_2 \rightarrow 2\,H_2O$ *slow*

unit5mcq

105. $O_3(g) + O(g) \rightarrow 2\ O_2(g)$

The decomposition of $O_3(g)$ in the upper atmosphere is represented by the equation above. The potential energy in the upper atmosphere is diagram for the decomposition of $O_3(g)$ in the presence and absence of $NO(g)$ is given below.



Which of the following mechanisms for the catalyzed reaction is consistent with the equation and diagram above?

(A) $2\ O_3(g) + 2\ NO(g) \rightarrow 4\ O_2(g) + N_2(g)$ *slow*

(B) $O_3(g) + NO(g) \rightarrow NO_2(g) + O_2(g)$ *slow*
$NO_2(g) + O(g) \rightarrow NO(g) + O_2(g)$ *fast* ✓

(C) $NO_2(g) + O_3(g) \rightarrow NO(g) + 2\ O_2(g)$ *slow*
$NO(g) + O(g) \rightarrow NO_2(g)$ *fast*

(D) $NO_2(g) + O(g) \rightarrow NO_3(g)$ *slow*
$NO_3(g) + O_3(g) \rightarrow NO_2(g) + 2\ O_2(g)$ *fast*

106. $N_2O(g) + CO(g) \rightarrow N_2(g) + CO_2(g)$

The rate of the reaction represented above increases significantly in the presence of $Pd(s)$. Which of the following best explains this observation?

(A) Pd increases the activation energy of the reaction.

(B) Pd absorbs the heat produced in the reaction.

(C) One of the reactants binds on the surface of Pd, which introduces an alternative reaction pathway with a lower activation energy. ✓

(D) One of the products binds on the surface of Pd, which increases the reaction rate by decreasing the concentration of products in the mixture.

unit5mcq

107.

Step 1: $\mathrm{NO_2}(g) + \mathrm{F_2}(g) \rightleftarrows \mathrm{NO_2F_2}(g)$ (*fast*)

Step 2: $\mathrm{NO_2F_2}(g) \rightarrow \mathrm{NO_2F}(g) + \mathrm{F}(g)$ (*slow*)

Step 3: $\mathrm{F}(g) + \mathrm{NO_2}(g) \rightarrow \mathrm{NO_2F}(g)$ (*fast*)

A proposed mechanism for the chemical reaction $2\ \mathrm{NO_2}(g) + \mathrm{F_2}(g) \rightarrow 2\ \mathrm{NO_2F}(g)$is shown above. Which of the following rate laws is consistent with this mechanism?

(A) $\text{Rate} = k[\mathrm{NO_2}][\mathrm{F_2}]$ ✓

(B) $\text{Rate} = k[\mathrm{NO_2}]^2[\mathrm{F_2}]$

(C) $\text{Rate} = k[\mathrm{NO_2}][\mathrm{F_2}]^2$

(D) $\text{Rate} = k[\mathrm{NO_2}]^2[\mathrm{F_2}]^2$

108. Step 1. $N_2H_2O_2 \rightleftarrows N_2HO_2^- + H^+$ (fast equilibrium)

Step 2. $N_2HO_2^- \rightarrow N_2O + OH^-$ (slow)

Step 3. $H^+ + OH^- \rightarrow H_2O$ (fast)

Nitramide, $N_2H_2O_2$, decomposes slowly in aqueous solution. This decomposition is believed to occur according to the reaction mechanism above. The rate law for the decomposition of nitramide that is consistent with this mechanism is given by which of the following?

(A) Rate = $k[N_2H_2O_2]$

(B) Rate = $k[N_2H_2O_2]\ [H^+]$

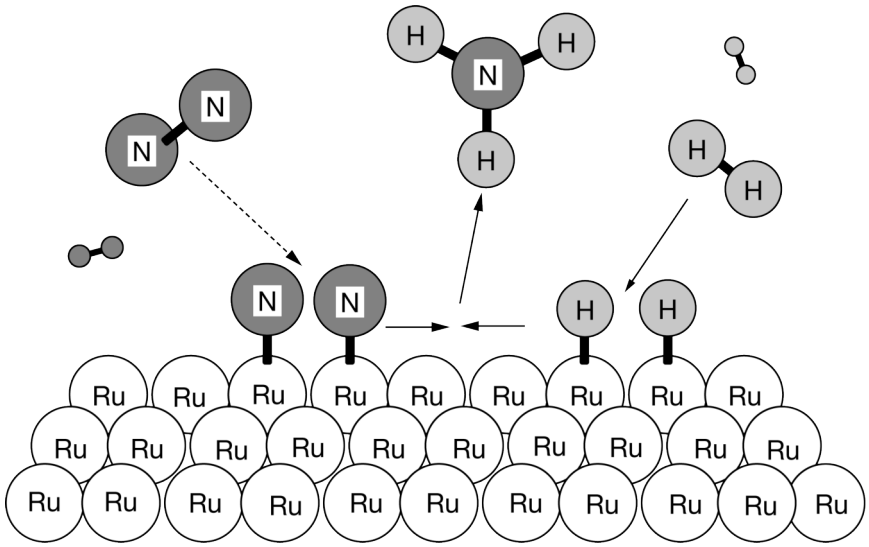
(C) Rate = $k\frac{[N_2H_2O_2]}{[H^+]}$ ✓

(D) Rate = $k\frac{[N_2H_2O_2]}{[N_2HO_2]}$

(E) Rate = $k[N_2H_2O_2]\ [OH^-]$

unit5mcq

109.



The diagram above illustrates how the reaction $N_2(g) + 3\ H_2(g) \rightarrow 2\ NH_3(g)$ occurs on the surface of $Ru(s)$. The rate of this reaction is determined by the amount of energy required to break the bond in N_2. This bond is weakened when N_2 is adsorbed on $Ru(s)$. Based on this information, which of the following provides the best reason for the use of $Ru(s)$ for the synthesis of NH_3 ?

(A) It promotes the proper orientation of N and H atoms to form new $N - H$ bonds.

(B) It provides a reaction path with a lower activation energy. ✓

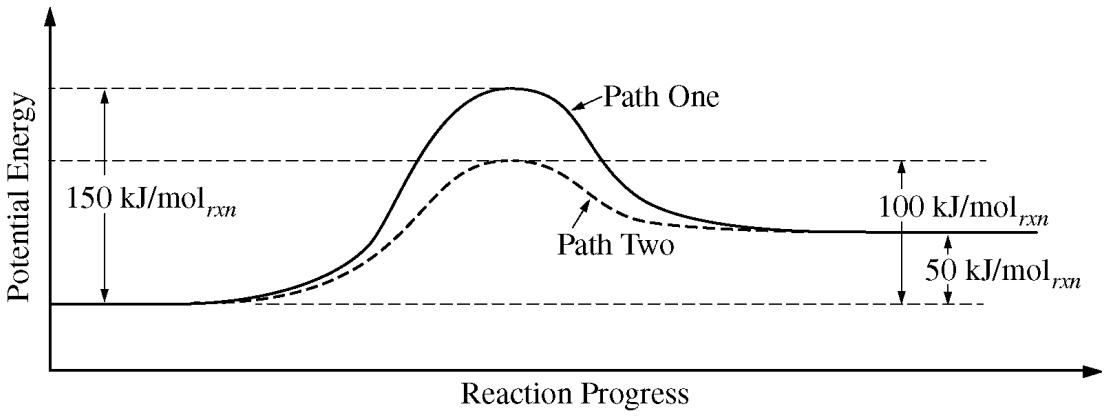
(C) It forms a bond between Ru and N that is stronger than the bond in N_2.

(D) It decreases the frequency of collisions between $N_2(g)$ and $H_2(g)$.

The following questions relate to the below information.

$$XY_2 \rightarrow X + Y_2$$

The equation above represents the decomposition of a compound XY_2. The diagram below shows two reaction profiles (path one and path two) for the decomposition of XY_2.



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unit5mcq

110. Which of the following most likely accounts for the difference between reaction path one and reaction path two?

(A) A higher temperature in path one

(B) A higher temperature in path two

(C) The presence of a catalyst in path one

(D) The presence of a catalyst in path two ✓

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111. $2\ H_2(g) + 2\ NO(g) \rightarrow N_2(g) + 2\ H_2O(g)$

For the chemical reaction represented above, the following mechanism is proposed.

Step 1:	$2\ NO(g) \rightleftharpoons N_2O_2(g)$	(fast equilibrium)
Step 2:	$N_2O_2(g) + H_2(g) \rightarrow N_2O(g) + H_2O(g)$	(slow)
Step 3:	$N_2O(g) + H_2(g) \rightarrow N_2(g) + H_2O(g)$	(fast)

Which of the following initial rate law expressions is consistent with this proposed mechanism?

(A) $\text{Rate} = k\left[H_2\right]^2\left[NO\right]^2$

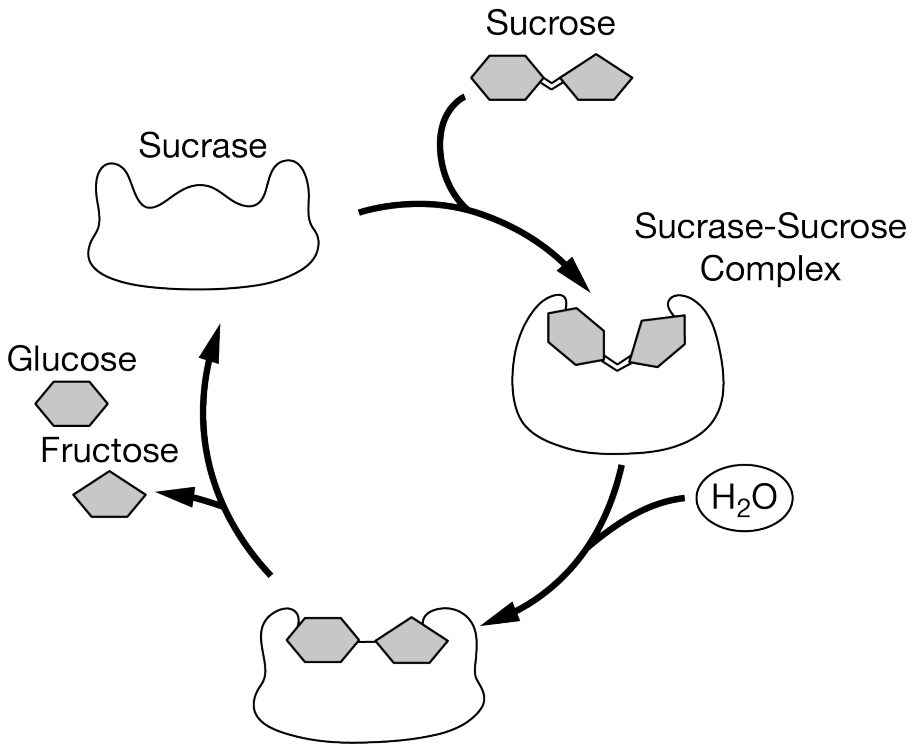
(B) $\text{Rate} = k[NO]^2$

(C) $\text{Rate} = k[H_2][NO]^2$ ✓

(D) $\text{Rate} = k[H_2][N_2O]^2$

unit5mcq

112.



$C_{12}H_{22}O_{11} + H_2O \rightarrow 2\ C_6H_{12}O_6$

The hydrolysis of sucrose is represented by the chemical equation above. This reaction is extremely slow in aqueous solution. However, when sucrase is added as shown in the diagram, the rate of the reaction is about 6,000,000 times faster. Based on this information, which of the following best explains the large increase in the rate of hydrolysis that occurs with the addition of sucrase?

(A) The reaction proceeds through a different reaction path with a lower activation energy in which the sucrase-sucrose complex is formed as an intermediate. ✓

(B) The addition of the sucrase reduces the number of effective collisions between the water and sucrose molecules.

(C) The reaction proceeds through a different reaction path that is independent of temperature and concentration.

(D) The addition of the sucrase decreases the number of sucrose molecules that have the minimum energy required to overcome the activation energy barrier.

113. The role of a catalyst in a chemical reaction is to

(A) decrease the amount of reactants that must be used

(B) lower the activation energy for the reaction ✓

(C) supply the activation energy required for the reaction to proceed

(D) increase the amounts of products formed at equilibrium

(E) increase the entropy change for the reaction